ISLAMIC UNIVERSITY OF TECHNOLOGY
Computer Science and Engineering (CSE)
Chem 4241: Chemistry

Chemical Bonding

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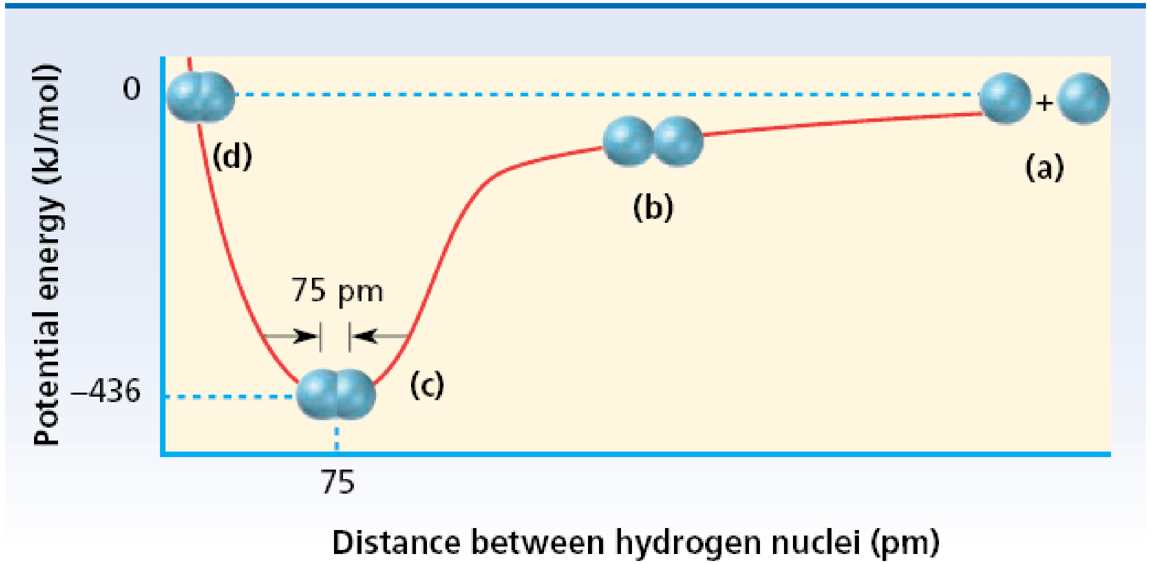
Chemical Bond

A *chemical bond* is a strong attractive force that exists between certain atoms in a substance.

Why do atoms form chemical bonds???

- As independent particles, atoms are at relatively high potential energy, less stable.
- Nature always favors the arrangements in which potential energy is minimized, more stable.
- By bonding with each other, the potential energy decreased, thereby creating more stable arrangements of matter.

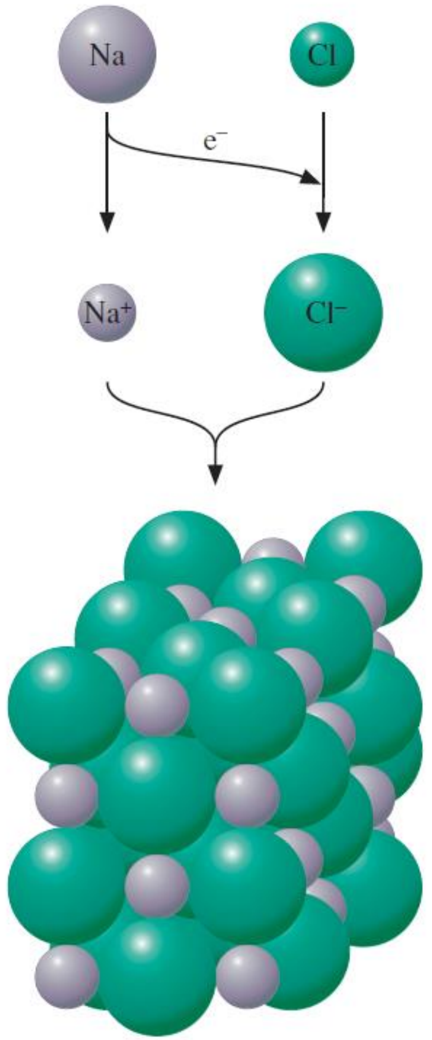


Types of Chemical Bond

There are three different types of bonds recognized by chemists :

1. Ionic bond
2. Covalent bond
3. Coordinate covalent bond

There is fourth type of bond, called metallic bond, we will discuss later.



Ionic bond

An *ionic bond* is a chemical bond formed by the electrostatic attraction between positive and negative ions. The bond forms between two atoms when one or more electrons are transferred from the valence shell of one atom to the valence shell of the other. The atom that loses electrons becomes a cation (positive ion), and the atom that gains electrons becomes an anion (negative ion).

$$\mathrm{Na([Ne]3}s^1) + \mathrm{Cl([Ne]3}s^2 3p^5) \longrightarrow \mathrm{Na^+([Ne])} + \mathrm{Cl^-([Ne]3}s^2 3p^6)$$

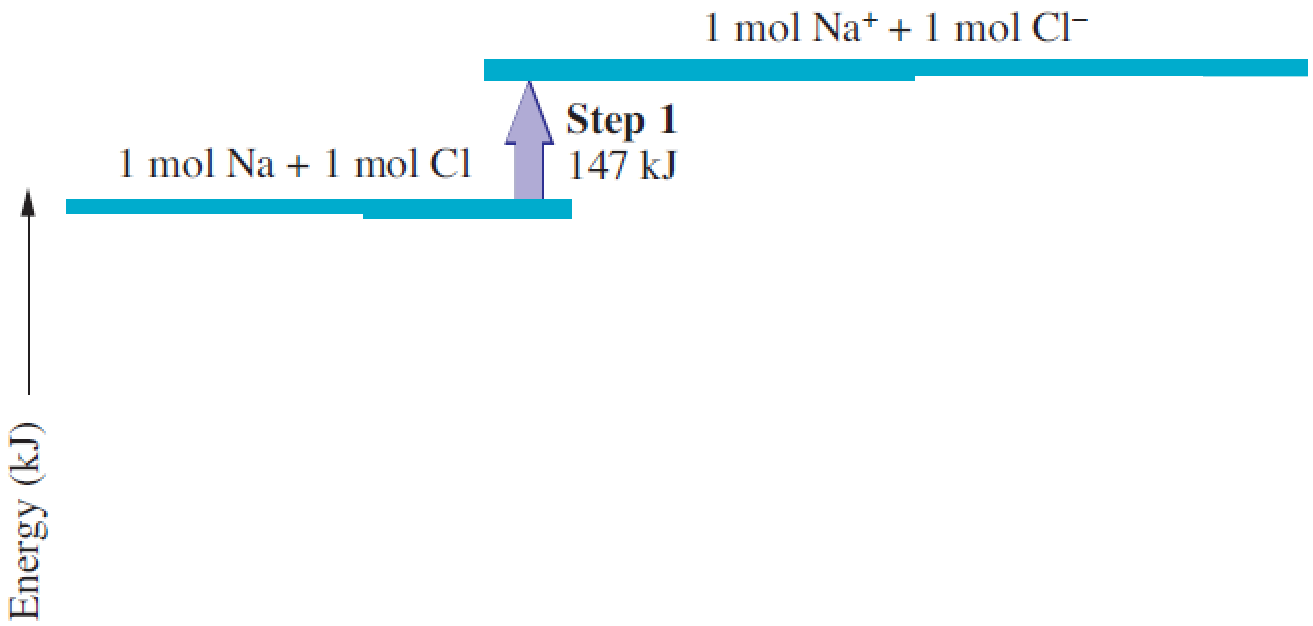
Within the sodium chloride crystal, NaCl, every Na^+ ion is surrounded by six Cl^- ions, and every Cl^- ion by six Na^+ ions.

Energy Involved in Ionic Bonding

If atoms come together and bond, there should be a net decrease in energy, because the bonded state should be more stable and therefore at a lower energy level.

$Na\ (1s^2 2s^2 2p^6 3s^1) \rightarrow Na^+ (1s^2 2s^2 2p^6) + e^-$ $\Delta H^0 = 496$ kJ/mol

$Cl\ (1s^2 2s^2 2p^6 3s^1 3p^5) + e^- \rightarrow Cl^- (1s^2 2s^2 2p^6 3s^2 3p^6)$ $\Delta H^0 = -349$ kJ/mol



The *lattice energy* is the change in energy that occurs when an ionic solid is separated into isolated ions in the gas phase.
For sodium chloride, the process is
$NaCl(s) \rightarrow Na^+(g) + Cl^-(g)$

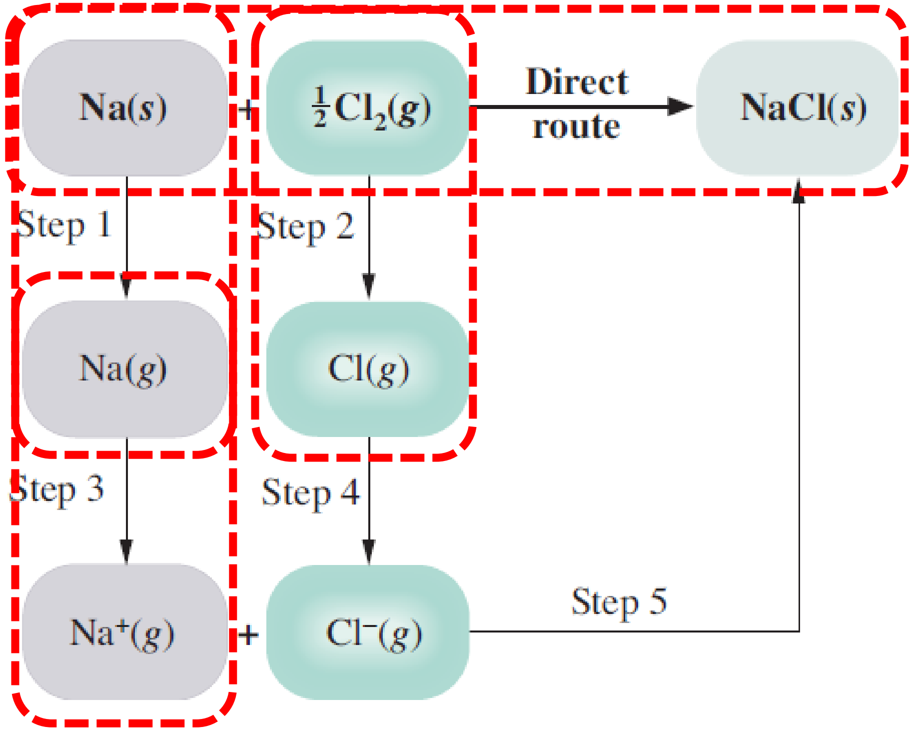
Lattice Energies from the Born–Haber Cycle

To obtain the lattice energy of NaCl, let the solid sodium chloride being formed from the elements by two different routes, as shown in Figure. In one route, NaCl(s) is formed directly from the elements, Na(s) and $1/2Cl_2$(g).

$$Na(s) + \frac{1}{2}Cl_2(g) \rightarrow NaCl(s) \qquad \Delta H_f^0 = -411\ kJ/mol$$

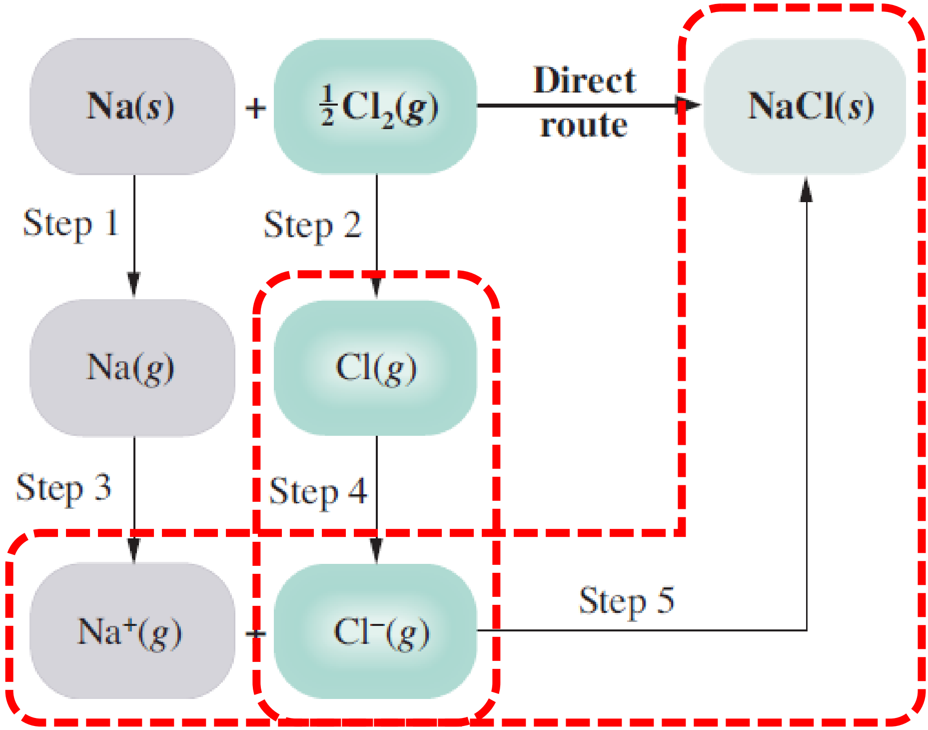
The second route consists of five steps:

1. *Sublimation of sodium.* Metallic sodium is vaporized to a gas of sodium atoms. (*Sublimation* is the transformation of a solid to a gas.) The enthalpy change for this process, measured experimentally, is 108 kJ per mole of sodium.
2. *Dissociation of chlorine.* Chlorine molecules are dissociated to atoms. The enthalpy change for this equals the Cl-Cl bond dissociation energy, which is 240 kJ per mole of bonds, or 120 kJ per mole of Cl atoms.
3. *Ionization of sodium.* Sodium atoms are ionized to Na^+ ions. The enthalpy change is essentially the ionization energy of atomic sodium, which equals 496 kJ per mole of Na.



Lattice Energies from the Born–Haber Cycle

4. *Formation of chloride ion.* The electrons from the ionization of sodium atoms are transferred to chlorine atoms. The enthalpy change for this is the electron affinity of atomic chlorine, which equals -349 kJ per mole of Cl atoms.
5. *Formation of NaCl(s) from ions.* The ions Na^+ and Cl^- formed in Steps 3 and 4 combine to give solid sodium chloride. Because this process is just the reverse of the one corresponding to the lattice energy (breaking the solid into ions), the enthalpy change is the negative of the lattice energy. If we let *U* be the lattice energy, the enthalpy change for Step 5 is -*U*.



Lattice Energies from the Born–Haber Cycle

Let's write these five steps and add them, and also add the corresponding enthalpy changes, following Hess's law.

$$
\begin{array}{lcll}
Na(s) & \longrightarrow & \cancel{Na(g)} & \Delta H_1 = 108\ kJ \\
\frac{1}{2}Cl_2(g) & \longrightarrow & \cancel{Cl(g)} & \Delta H_2 = 120\ kJ \\
\cancel{Na(g)} & \longrightarrow & \cancel{Na^+(g)} + \cancel{e^-(g)} & \Delta H_3 = 496\ kJ \\
\cancel{Cl(g)} + \cancel{e^-(g)} & \longrightarrow & \cancel{Cl^-(g)} & \Delta H_4 = -349\ kJ \\
\cancel{Na^+(g)} + \cancel{Cl^-(g)} & \longrightarrow & NaCl(s) & \Delta H_5 = -U \\
\hline
\boxed{Na(s) + \frac{1}{2}Cl_2(g) \longrightarrow NaCl(s)} & & & \Delta H_f^\circ = 375\ kJ - U
\end{array}
$$

The final equation is simply the formation reaction for NaCl(*s*). The enthalpy of formation has been determined calorimetrically and equals -411 kJ.

Equating these two values, we get

$$375kJ - U = -411\ kJ$$

Solving for *U* yields the lattice energy of NaCl:

$$U = (375+411)\ kJ = 786\ kJ$$

Factors Favoring The Formation of Ionic Bonds

1. **Number of valence electrons:** The atom forming the cation should possess 1, 2 or 3 valence electrons, while the other atom forming the anion should have 5, 6 or 7 valence electrons. The elements of group IA, IIA and IIIA satisfy this condition for cation and those of groups VA, VIA, and VIIA satisfy this condition for anion.
2. **Net lowering of Energy:** To form a stable ionic compound, there must be a net lowering of energy. In other words, energy must be released as a result of the electron transfer and formation of ionic compound by the following steps :
   a) The removal of electron from an atom requires energy, which is described by ionization energy (IE). Lower the ionization energy greater will be the tendency of the metal atom to form cation and hence greater will be the ease of formation of ionic bond.
   b) The addition of electron to an atom releases energy, which is described by electron affinity (EA). The higher the electron affinity, the more energy is released, resulting in a more stable anion formation which favors the formation of ionic bond.
   c) The electrostatic attraction between cations and anions in the solid compound releases energy, which is related to the lattice energy. Greater lattice energy corresponds to a stronger ionic bond.

Factors Favoring The Formation of Ionic Bonds

The value of lattice energy depends on two main factors:

- **Size of the ions:** The force of attraction between cations and anions is inversely proportional to the square of the distance between them. Therefore, smaller ion sizes result in greater attraction and higher lattice energy.
- **Charge on ions:** The strength of the ionic bond is directly proportional to the charge on the ions. Greater charges on the ions lead to stronger attraction and higher lattice energy.

If the energy released in steps (b) and (c) is greater than the energy consumed in step (a), the overall process of electron transfer and formation of the ionic compound results in a net release of energy, contributing to the stability of the ionic bond.

3. **Electronegativity difference of A and B**

The formation of an ionic bond typically requires a significant difference in electronegativity between the atoms involved, generally greater than 2. This large difference ensures that one atom strongly attracts electrons, becoming a negatively charged ion (anion), while the other atom loses electrons to become a positively charged ion (cation). For example, sodium (Na) has an electronegativity of 0.9, while chlorine (Cl) has an electronegativity of 3.0. The difference in electronegativity between Na and Cl (3.0 - 0.9 = 2.1) is sufficient for them to form an ionic bond.

Characteristics of Ionic Compounds

- **Solids at Room Temperature**

Due to the strong electrostatic attraction between oppositely charged ions, they are held firmly in place within the crystal lattice. This strong bond prevents them from moving freely, a characteristic of the liquid state, thereby causing the substance to exist as a solid at room temperature.

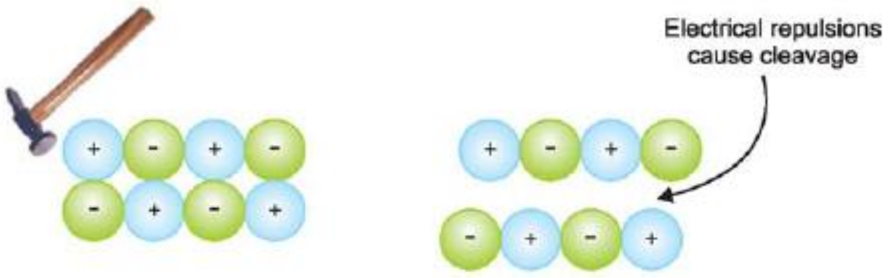
- **High Melting Points**

Ionic compounds typically exhibit high melting and boiling points. This is due to the strong electrostatic forces of attraction between the positively and negatively charged ions in the lattice structure. At high temperatures, the ions gain enough kinetic energy to overcome these attractive forces, allowing them to break free from their fixed positions in the lattice and move more freely, as seen in a liquid state.

- **Hard and brittle**

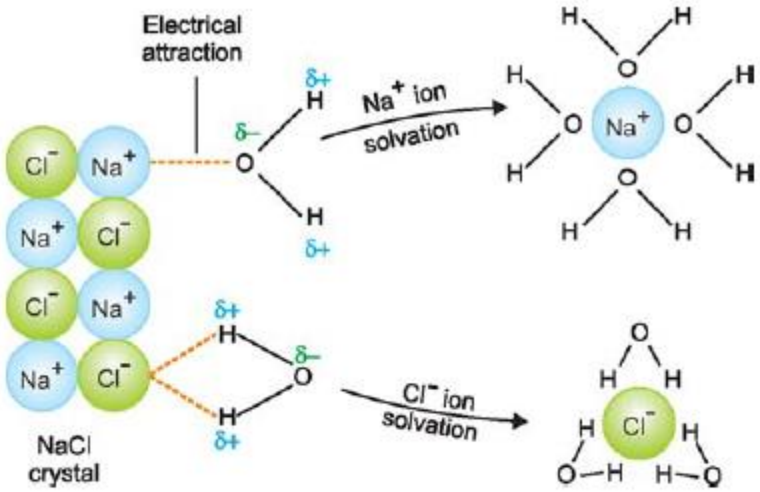
The crystals of ionic substances are hard and brittle due to the strong electrostatic forces that hold each ion in its designated position within the crystal lattice. These crystals are composed of layers of positively charged ions and negatively charged ions arranged alternately so that ions of opposite charges in adjacent layers align. When an external force is applied to a layer of ions relative to the next layer, even a slight shift can bring like-charged ions into proximity. This proximity leads to repulsion between the similarly charged ions, causing the crystal to fracture along planes of weakness. This phenomenon is known as crystal cleavage.

Characteristics of Ionic Compounds



- **Soluble in water**

When a crystal of an ionic substance is immersed in water, the polar water molecules exert an electrostatic pull that detaches the positively charged (+) and negatively charged (–) ions from the crystal lattice. These liberated ions become surrounded by water molecules, allowing them to exist independently and causing the ionic substance to dissolve in water. Conversely, non-polar solvents such as benzene (C_6H_6) and hexane (C_6H_{14}) are unable to dissolve ionic compounds due to the absence of the necessary electrostatic interactions.



Characteristics of Ionic Compounds

- **Conductors of electricity**

Solid ionic compounds are poor conductors of electricity because the ions are rigidly fixed in their positions within the crystal lattice, preventing the flow of electrical charge. However, in the molten state or when dissolved in water, the ionic bonds break, and the ions become free to move. As a result, molten ionic compounds or their aqueous solutions can conduct electricity when placed in an electrolytic cell.

- **Do not exhibit isomerism**

The ionic bond, formed by the electrostatic attraction between oppositely charged ions, is non-rigid and non-directional. Consequently, ionic compounds lack the structural flexibility required for exhibiting isomerism.

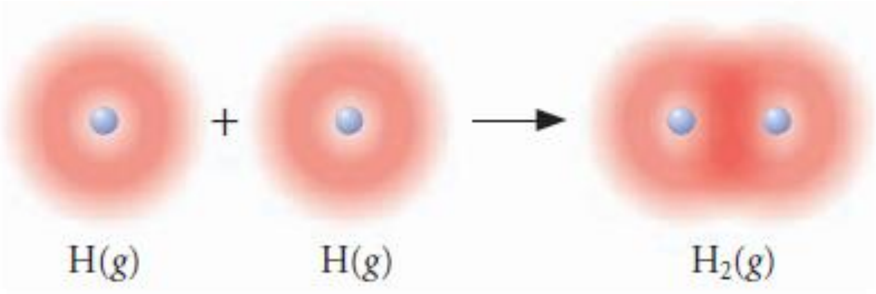
- **Ionic reactions are fast**

Ionic compounds give reactions between ions, and these are very fast.

Covalent Bond

Covalent bond is a chemical bond formed by the sharing of a pair of electrons between atoms.

$$H\cdot + \cdot H \longrightarrow H:H \qquad H:H$$



$$H\cdot + \cdot\overset{\cdot\cdot}{\underset{\cdot\cdot}{Cl}}: \longrightarrow H:\overset{\cdot\cdot}{\underset{\cdot\cdot}{Cl}}:$$

Factors Favoring The Formation of Covalent Bonds

- **Number of valence electrons**

  Each of the atom *A* and *B* should have 5, 6 or 7 valence electrons so that both achieve the stable octet by sharing 3, 2 or 1 electron-pair. H has one electron in the valence shell and attains duplet. The non-metals of groups VA, VIA and VIIA respectively satisfy this condition.

- **Equal electronegativity**

  The atom *A* will not transfer electrons to *B* if both have equal (or nearly equal) electronegativity, and hence electron sharing will take place. This can be strictly possible only if both the atoms are of the same element.

- **Equal sharing of electrons**

  The atoms *A* and *B* should have equal (or nearly equal) electron affinity so that they attract the bonding electron pair equally. Thus equal sharing of electrons will form a nonpolar covalent bond. Of course, precisely equal sharing of electrons will not ordinarily occur except when atoms *A* and *B* are atoms of the same element, for no two elements have exactly the same electron affinity.

Factors Favoring The Formation of Covalent Bonds

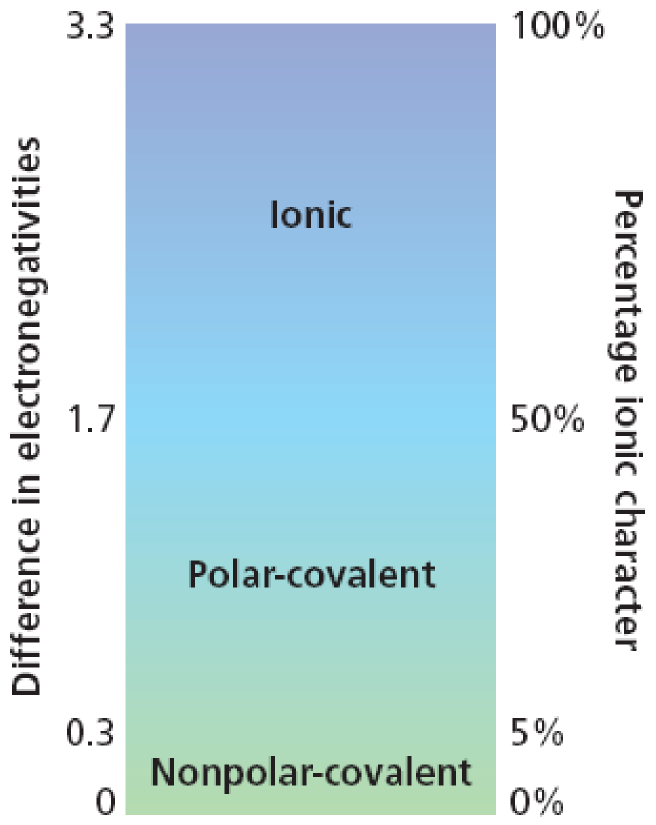
- Extent of bonding between atoms of 2 elements being ionic or covalent depends on difference in electronegativity between two bonded atoms.

- Fluorine's EN = 4.0, Cesium's EN = 0.7

  4.0-0.7 = 3.3

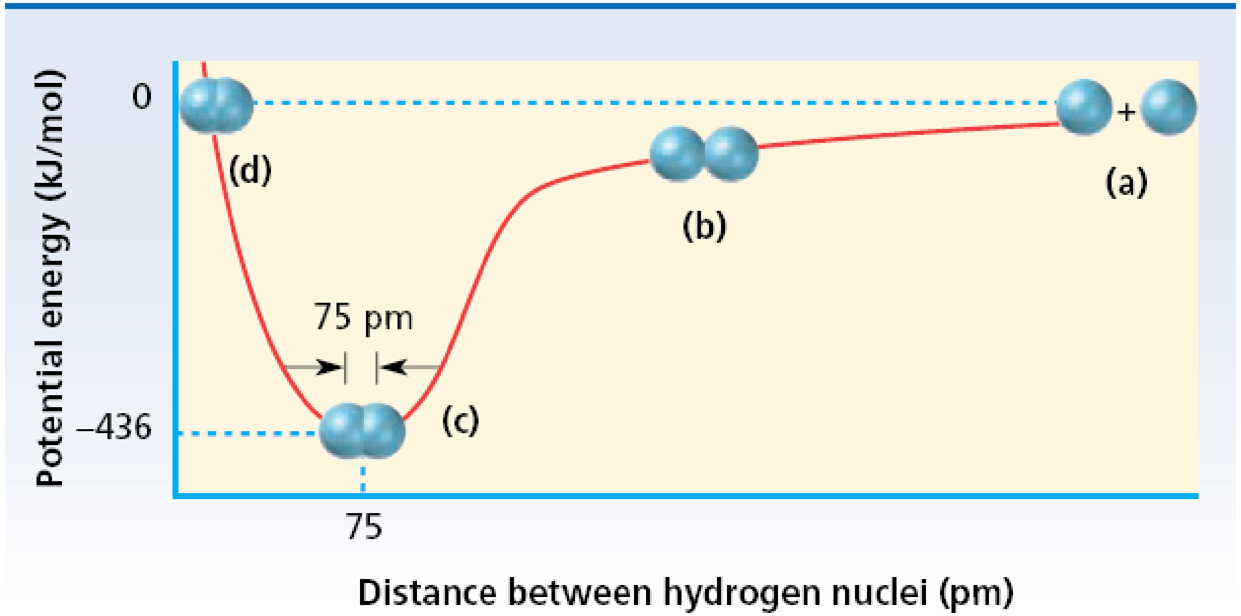
- According to table, F-Cs is ionic

- The greater the difference, the more ionic the bond



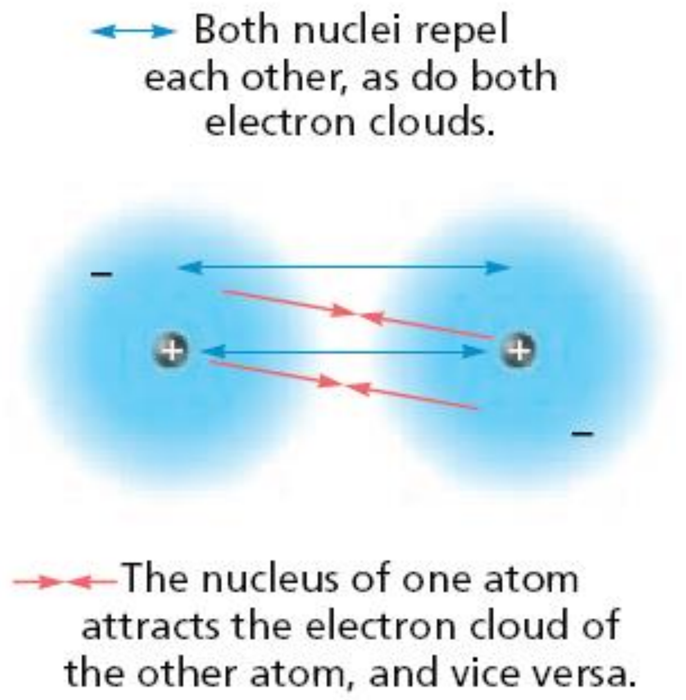
Energy Involved in Covalent Bonding

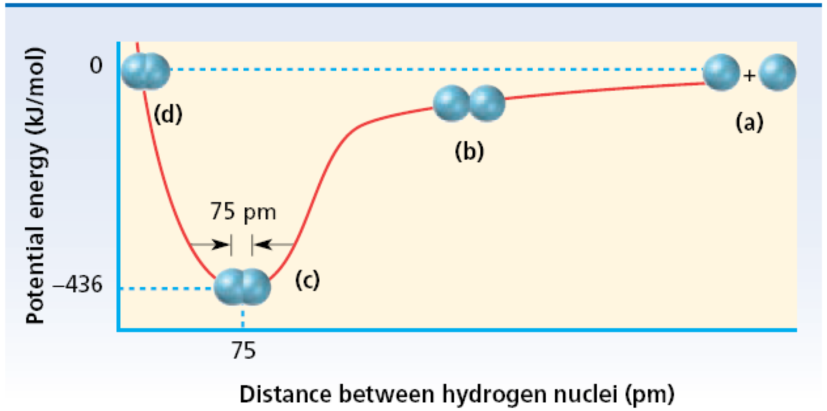
- Bonded atoms have lower potential energy than unbonded atoms
- At large distance atoms don't influence each other, and potential energy set at 0 at this stage.
- Consider formation of hydrogen molecules(H_2); Each H has (+) proton and nucleus surrounded by (-) electron
- As atoms near each other, charged particles start to interact
- Approaching nuclei and electrons are attracted to each other and decrease in total potential energy



Energy Involved in Covalent Bonding

- The amount of attraction/repulsion depends on how close the atoms are to each other
- When atoms first "see" each other, electron-proton attraction stronger than e-e or p-p repulsions
- So atoms drawn to each other and potential energy lowered
- Attractive force dominates until a distance is reached where repulsion equals attraction
- As long as energy stays close to minimum, they stay covalently bonded (Valley of the curve)
- Closer the atoms get, potential energy rises sharply
- Repulsion becomes greater than attraction





Characteristics of Covalent Compounds

While the atoms in a covalent molecule are firmly held by the shared electron pair, the individual molecules are attracted to each other by weak van der Waals forces. Thus the molecules can be separated easily as not much energy is required to overcome the intermolecular attractions. This explains the general properties of covalent compounds.

- **Gases, liquids or solids at room temperature**

The covalent compounds are often gases, liquids or relatively soft solids under ordinary conditions. This is so because of the weak intermolecular forces between the molecules.

- **Low melting points and boiling points**

Covalent compounds have generally low melting points (or boiling points). The molecules are held together in the solid crystal lattice by weak forces. On application of heat, the molecules are readily pulled out and these then acquire kinetic energy for free movement as in a liquid.

Characteristics of Covalent Compounds

- **Neither hard nor brittle**

While the ionic compounds are hard and brittle, covalent compounds are neither hard nor brittle. There are weak forces holding the molecules in the solid crystal lattice. A molecular layer in the crystal easily slips relative to other adjacent layers and there are no 'forces of repulsion' like those in ionic compounds. Thus the crystals are easily broken and there is no sharp cleavage between the layers on application of external force.

- **Soluble in organic solvents**

In general, covalent compounds dissolve readily in nonpolar organic solvents (benzene, ether). The kinetic energy of the solvent molecules easily overcomes the weak intermolecular forces. Covalent compounds are insoluble in water. Some of them (alcohols, amines) dissolve in water due to hydrogen-bonding.

- **Non-conductors of electricity**

Since there are no (+) or (–) ions in covalent molecules, the covalent compounds in the molten or solution form are incapable of conducting electricity.

Characteristics of Covalent Compounds

- **Exhibit Isomerism**

Covalent bonds are rigid and directional, the atoms being held together by shared electron pair and not by electrical lines of force. This affords opportunity for various spatial arrangements and covalent compounds exhibit stereoisomerism.

- **Molecular reactions**

The covalent compounds give reactions where the molecule as a whole undergoes a change. Since there are no strong electrical forces to speed up the reaction between molecules, these reactions are slow.

Comparison of Ionic And Covalent Bonds

Ionic Bond	Covalent Bond
1. Formed by transfer of electrons from a metal to a non-metal atom. 2. Consists of electrostatic force between (+) and (–) ions. 3. Non-rigid and non-directional; cannot cause isomerism. **Properties of Compounds** 1. Solids at room temperature. 2. High melting and boiling points. 3. Hard and brittle. 4. Soluble in water but insoluble in organic solvents. 5. Conductors of electricity 6. Undergo ionic reactions which are fast.	1. Formed by sharing of electrons between nonmetal atoms. 2. Consists of a shared pair of electrons between atoms. 3. Rigid and directional : causes stereoisomerism. **Properties of Compounds** 1. Gases, liquids or soft solids. 2. Low melting and boiling points. 3. Soft, much readily broken 4. Insoluble in water but soluble in organic solvents. 5. Non-conductors of electricity. 6. Undergo molecular reactions which are slow.

Electronegativity and Covalent Bonds

Electronegativity is a measure of the ability of an atom in a molecule to draw bonding electrons to itself.

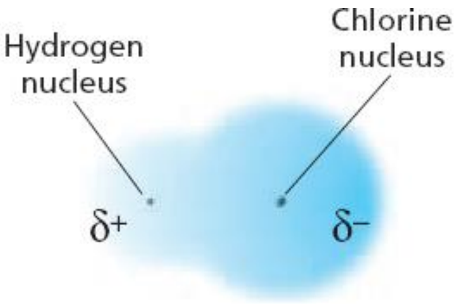
Nonpolar covalent Bond

- Nonpolar-covalent bond is a covalent bond in which the bonding electrons are shared equally by the bonded atoms, resulting in a balanced distribution of electrical charge
- 0-5% ionic character (0-0.3 EN difference) is nonpolar-covalent bond

(a) Nonpolar-covalent bond

Polar covalent bond

- Polar-covalent bond is a covalent bond in which the bonded atoms have an unequal attraction for the shared electrons
- Bonds that have significantly different ENs, electrons are more strongly attracted by more-EN atom
- These bonds are polar → *they have an uneven distribution of charge*

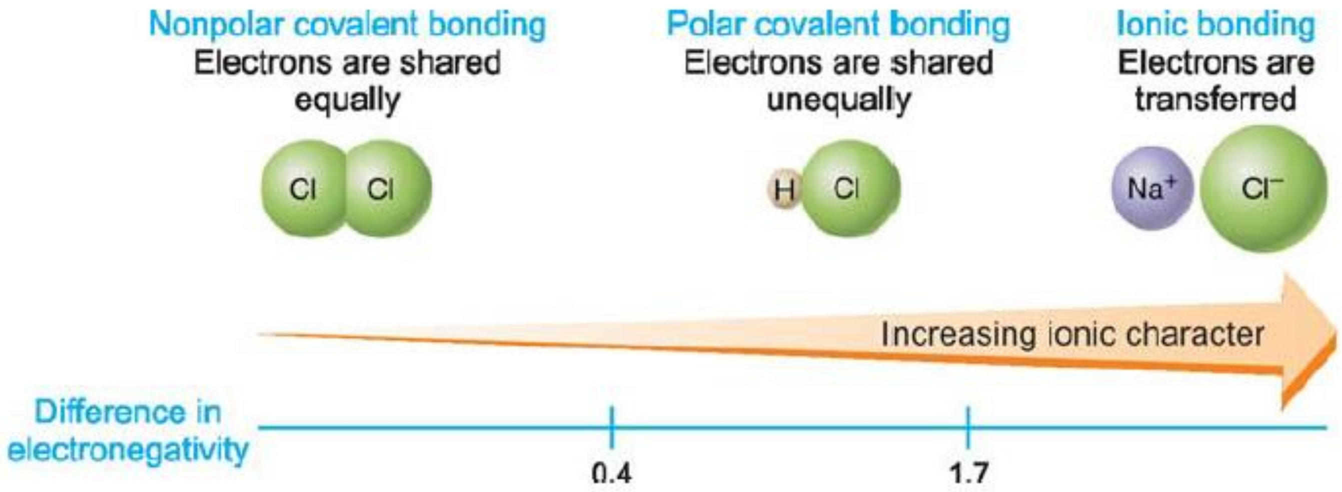


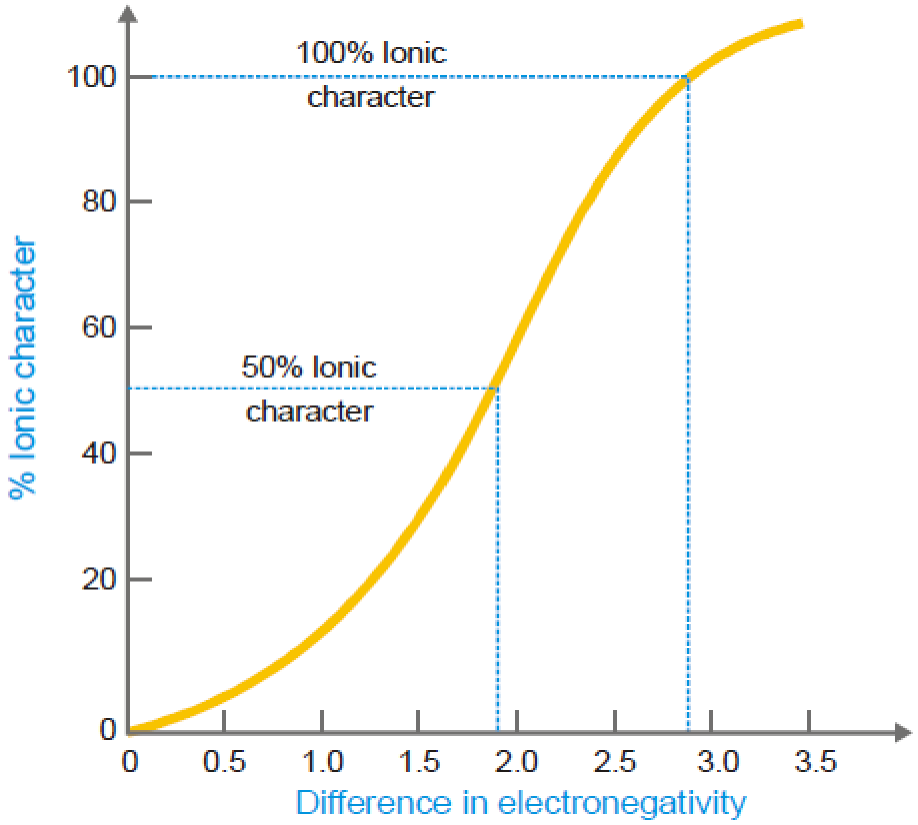
(b) Polar-covalent bond

The greater the difference of electronegativity between two atoms, greater the polarity

Electronegativity and Covalent Bonds

H=1/2(V+X-C+A)
L=H-X-D
Here, X= Monovalent atomic number
D





Electronegativity and Covalent Bonds

The percentage ionic character of a bond can be calculated by using the equation

$$\% \text{ ionic character} = 16\,[X_A - X_B] + 3.5\,[X_A - X_B]^2$$

This equation was given by **Hannay and Smith**.

Calculate the percentage ionic character of C–Cl bond in CCl_4 if the electronegativities of C and Cl are 3.5 and 3.0 respectively.

$$\% \text{ ionic character} = 16\,[X_A - X_B] + 3.5\,[X_A - X_B]^2$$

Give $X_A = 3.5$ and $X_B = 3.0$

$\therefore$ %a ionic character $= 16\,(3.5 - 3.0) + 3.5\,(3.5 - 3.0)^2$

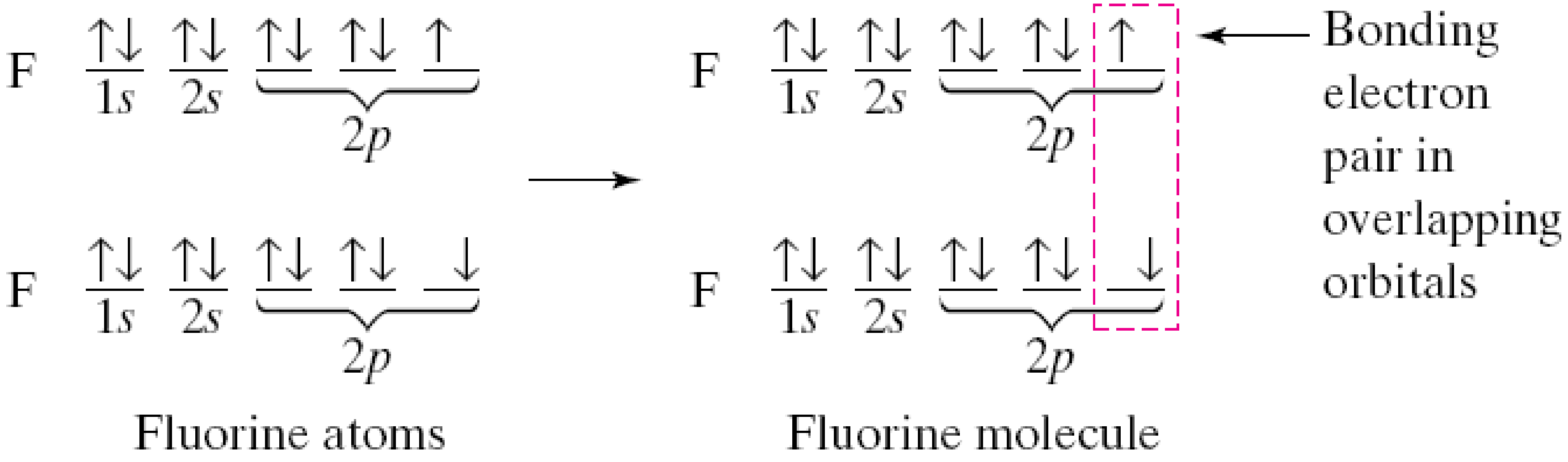
$= 8.0 + 0.875$

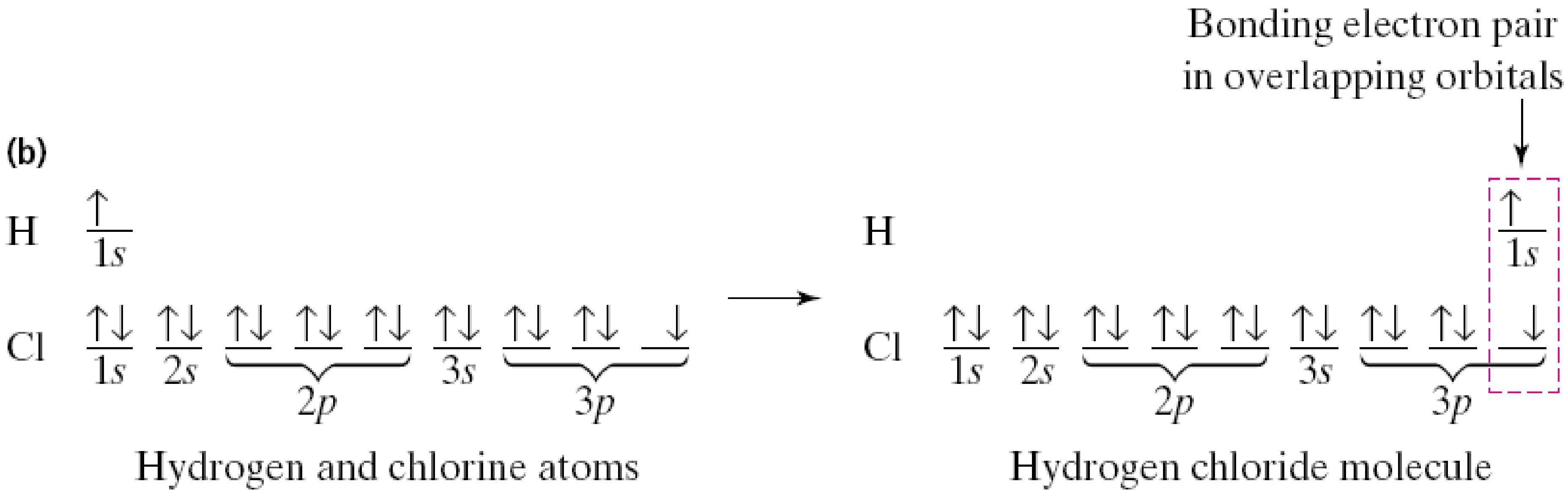
$= 8.875\%$

Octet Rule

- Noble-gas atoms have minimum energy existing on their own because of electronic configurations
- Outer orbitals completely full
- Other atoms full orbitals by transferring or sharing electrons
- Bond formation follows octet rule → chemical compounds tend to form so that each atom, by gaining, losing, or sharing electrons, has an octet of electrons in its highest occupied energy level

Octet Rule


F
1s
2s
2p
F
1s
2s
2p
Fluorine atoms
F
1s
2s
2p
F
1s
2s
2p
Fluorine molecule
Bonding electron pair in overlapping orbitals


(b)
H
1s
Cl
1s
2s
2p
3s
3p
Hydrogen and chlorine atoms
Bonding electron pair in overlapping orbitals
H
1s
Cl
1s
2s
2p
3s
3p
Hydrogen chloride molecule

Limitations of Octet Rule

- Most main-group elements form covalent bonds according to octet rule
- It is clear that octet rule is based upon the chemical inertness of noble gases. However, some noble gases (Xe, Kr) also combine with oxygen and fluorine to form a number of compounds like XeF_2, KrF_2, $XeOF_2$ etc. This theory does not account for the shape of molecules.
- Incomplete octet of the central atom e.g. H-H only 2 electrons
- Boron, B, has 3 valence electrons ($[He]2s^2 2p^1$) and boron tends to form bonds where it is surrounded by 6 e- (e- pairs)
- The Expanded Octet: Elements in and beyond the third period of the periodic table have, apart from 3s and 3p orbitals, 3d orbitals also available for bonding. In a number of compounds of these elements there are more than eight valence electrons around the central atom. This is termed as expanded octet.Examples- PF_5, SF_6, H_2SO_4

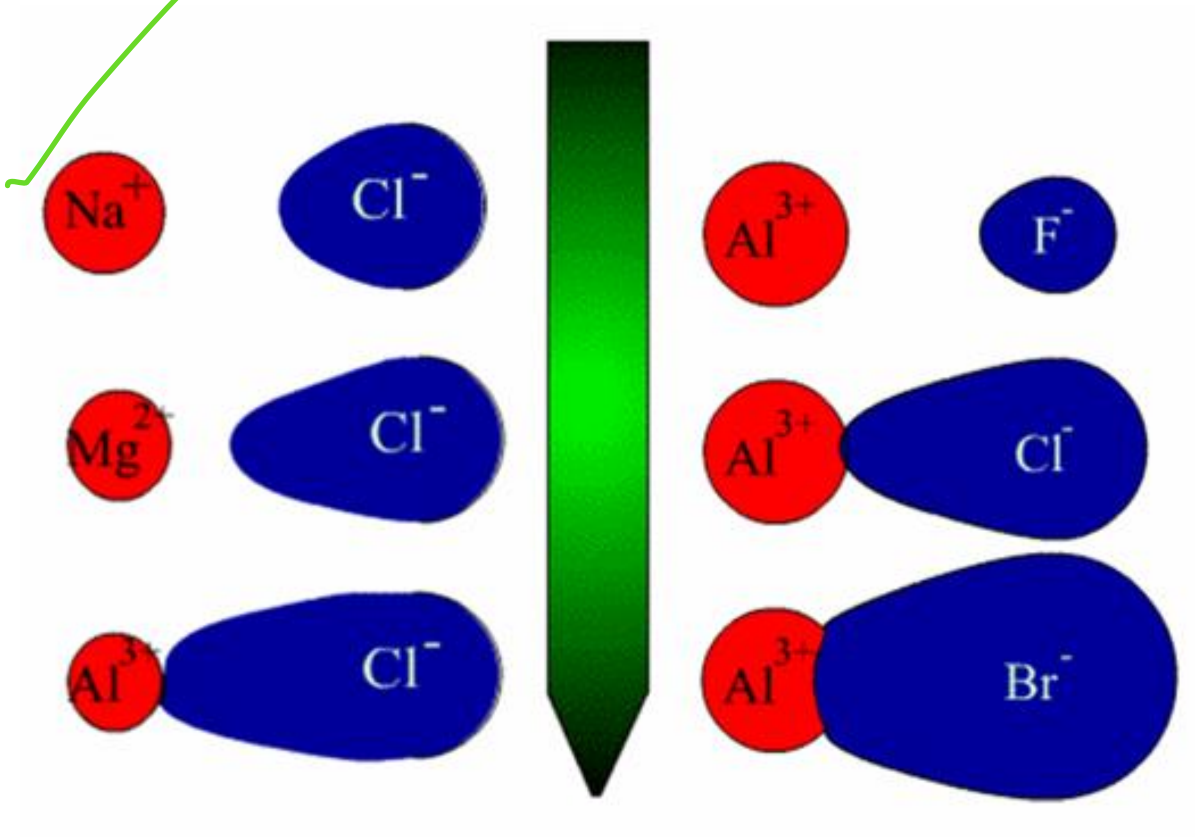
Fajan's Rules

The partial covalent character of ionic bonds was discussed by Fajan's in terms of the following rules:

- The smaller the size of the cation and the larger the size of the anion, the greater the covalent character of an ionic bond.
- The greater the charge on the cation, the greater the covalent character of the ionic bond.
- For cations of the same size and charge, the one, with electronic configuration (n-1)$d^n ns^0$, typical of transition metals, is more polarizing than the one with a noble gas configuration, ns^2 np^6, typical of alkali and alkaline earth metal cations.

The cation polarizes the anion, pulling the electronic charge toward itself and thereby increasing the electronic charge between the two. This is precisely what happens in a covalent bond, i.e., build-up of electron charge density between the nuclei. The polarizing power of the cation, the polarizability of the anion and the extent of distortion (polarization) of anion are the factors, which determine the percent covalent character of the ionic bond.

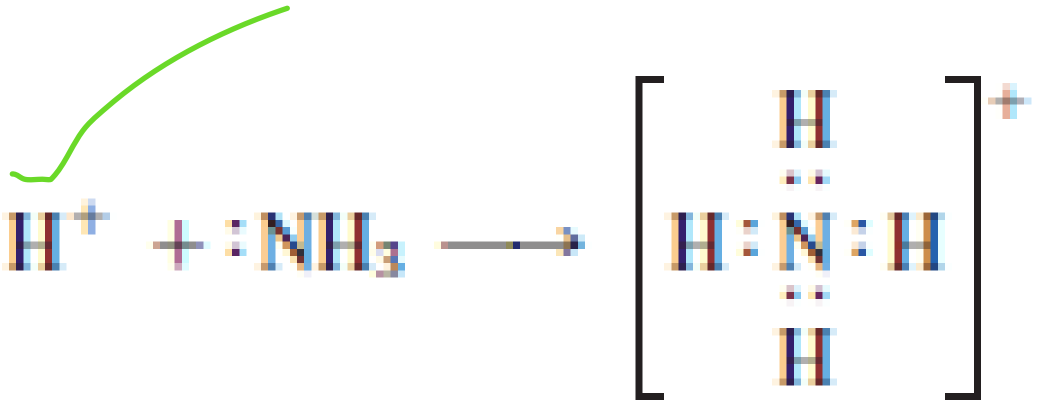
Fajan's Rules



Coordinate Covalent Bonds

A coordinate covalent bond is a bond formed when both electrons of the bond are donated by one atom but shared between two atoms.

For example, the formation of the ammonium ion, in which an electron pair on the N atom of NH_3 forms a bond with H^+.

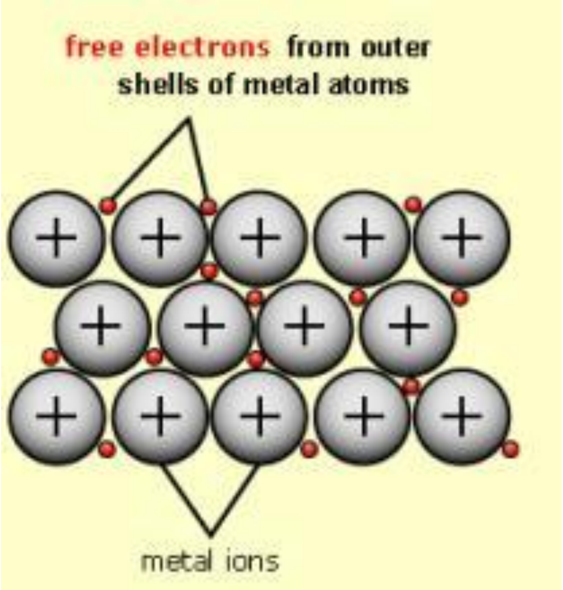


Metallic Bonding

Metallic bonding is the type of bonding found in metallic elements. This is the electrostatic force of attraction between positively charged ions and delocalized outer electrons (sea of electron).

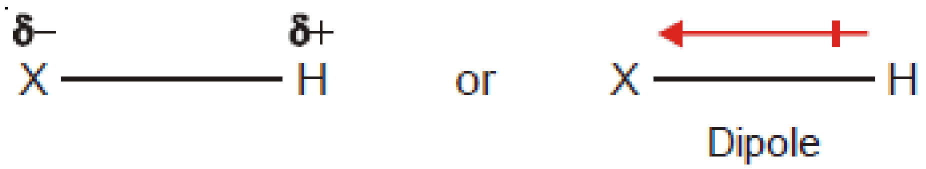
Metallic bonding refers to the interaction between the delocalized electrons and the metal nuclei.

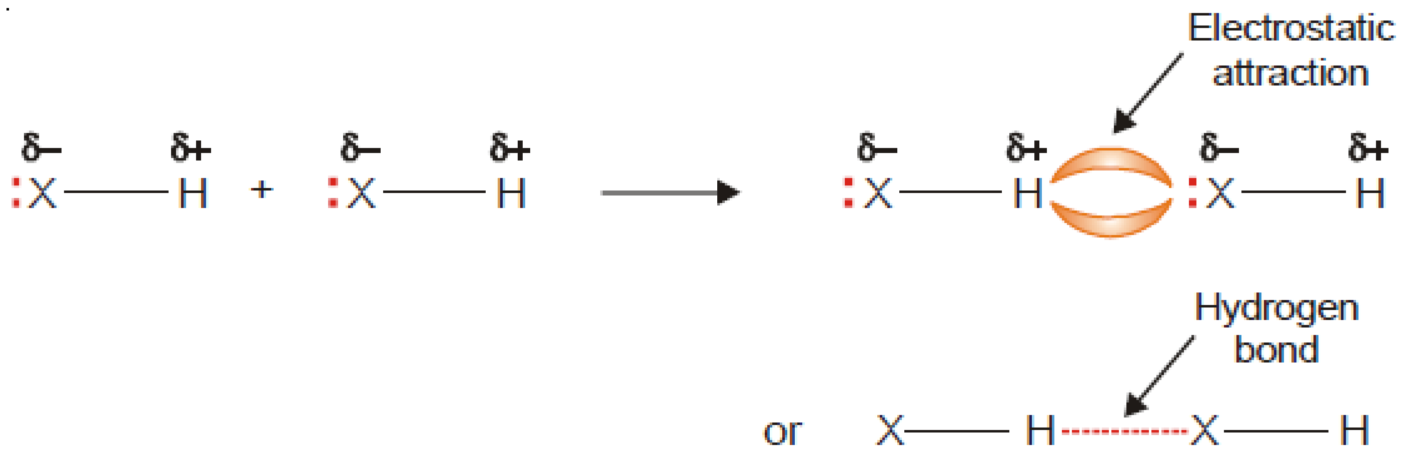
As the metal cations and the electrons are oppositely charged, they will be attracted to each other, and also to other metal cations. These electrostatic forces are called metallic bonds, and these are what hold the particles together in metals.



Hydrogen Bonding

A hydrogen bond is the attractive force between the hydrogen attached to an electronegative atom of one molecule and an electronegative atom of a different molecule. H atom attached to the electronegative atoms, oxygen, nitrogen or fluorine, are capable of forming H-bond.





Hydrogen Bonding

Types of hydrogen bonding

➢ Intermolecular hydrogen bonding

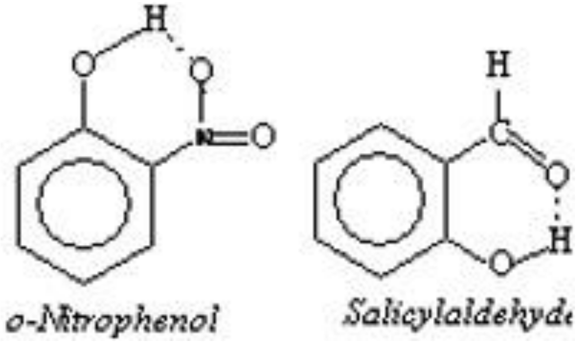
Present between two diff molecules

This type of hydrogen bonding involves electrostatic forces of attraction between hydrogen and electronegative element of two different molecules of the substance. Hydrogen bonding in molecules of HF, NH_3 , H_2O etc. are examples of intermolecular hydrogen bonding.

➢ Intramolecular hydrogen bonding

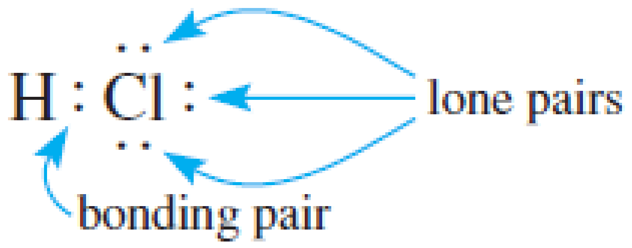
present between the same molecule

This type of bonding involves electrostatic forces of attraction between hydrogen and electronegative element both present in the same molecule of the substance. Examples o-nitrophenol and salicylaldehyde.



Lewis Electron-dot Formula

A formula using dots to represent valence electrons is called a Lewis electron-dot formula. An electron pair represented by a pair of dots in such a formula is either a bonding pair *(an electron pair shared between two atoms)* or a lone, or nonbonding, pair *(an electron pair that remains on one atom and is not shared)*. For example,



Bonding pairs are often represented by dashes rather than by pairs of dots.

How to Write Lewis Electron-dot Formula

Step 1: Calculate the total number of valence electrons for the molecule by summing the number of valence electrons (= group number) for each atom. For a polyatomic anion, add the number of negative charges to this total (For CO_3^{2-} add 2). For a polyatomic cation, subtract the number of positive charges from the total (For NH_4^+ subtract 1).

Step 2: Write the skeleton structure of the molecule or ion, connecting every bonded pair of atoms by a pair of dots or a dash. Usually, less electronegative atoms (central atom) are surrounded by more electronegative atoms.

Step 3: Distribute electrons to the atoms surrounding the central atom (or atoms) to satisfy the octet rule for these surrounding atoms.

Step 4: Distribute the remaining electrons as pairs to the central atom (or atoms).

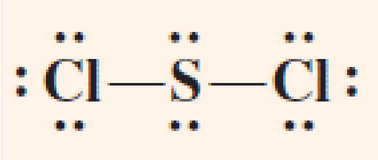
Step 5: If there are fewer than eight electrons on the central atom, this suggests that a multiple bond is present. To obtain a multiple bond, move one or two electron pairs (depending on whether the bond is to be double or triple) from a surrounding atom to the bond connecting the central atom. Atoms that often form multiple bonds are C, N, O, and S.

How to Write Lewis Electron-dot Formula

SCl_2:

Total number of valence electron: 20

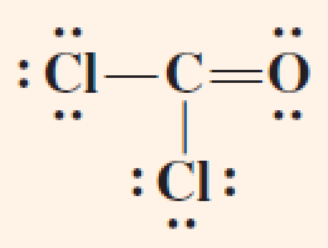
Central atom: S



$COCl_2$:

Total number of valence electron: 24

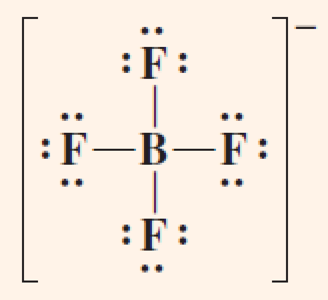
Central atom: C



BF_4^-:

Total number of valence electron: 31+1=32

Central atom: B

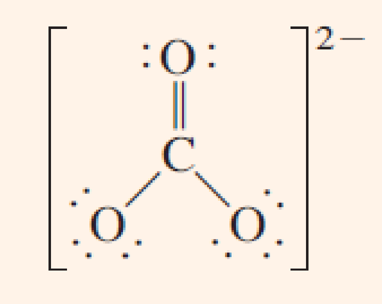


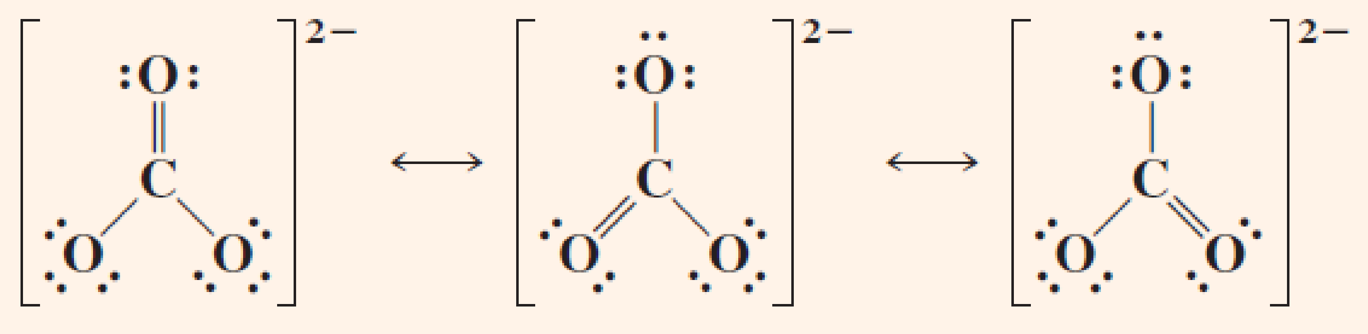
How to Write Lewis Electron-dot Formula

CO_3^{2-}:

Total number of valence electron: 22+2=24

Central atom: C





ClO_2^- CO_2 O_3 NO_3^- PF_5 XeF_4 SF_4 BF_3

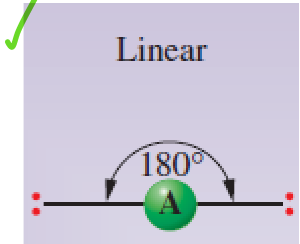
Valence Shell Electron Pair Repulsion Theory

The valence-shell electron-pair repulsion (VSEPR) model predicts the shapes of molecules and ions by assuming that the valence-shell electron pairs are arranged about each atom so that electron pairs are kept as far away from one another as possible, thus minimizing electron-pair repulsions.

The main postulates of VSEPR theory:

- The shape of a molecule depends upon the number of valence shell electron pairs (bonded or non-bonded) around the central atom.
- Pairs of electrons in the valence shell repel one another.
- These pairs of electrons tend to occupy such positions in space that minimise repulsion and thus maximise distance between them.
- The repulsive interaction of electron pairs decrease in the order: Lone pair (lp)–Lone pair (lp) > Lone pair (lp)–Bond pair (bp) > Bond pair (bp)–Bond pair (bp).
- The lone pairs are localised on the central atom, each bonded pair is shared between two atoms. As a result, the lone pair electrons in a molecule occupy more space as compared to the bonding pairs of electrons. This results in greater repulsion between lone pairs of electrons as compared to the lone pair – bond pair and bond pair – bond pair repulsions.

Valence Shell Electron Pair Repulsion Theory



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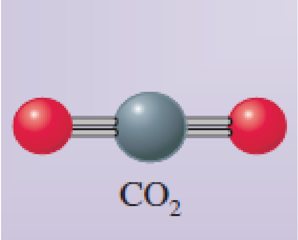

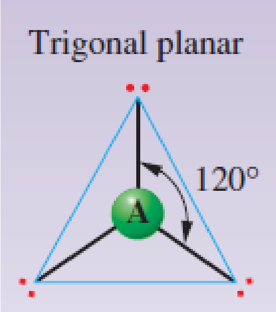

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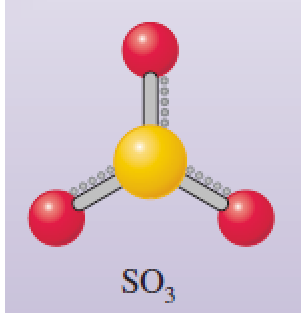

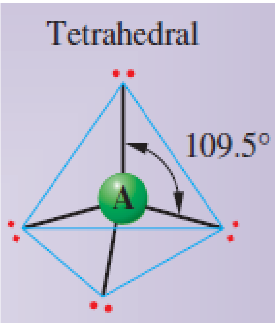

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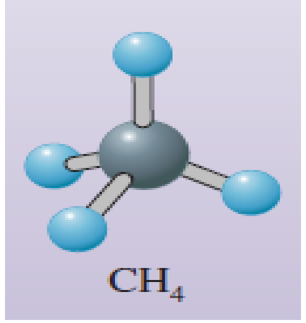

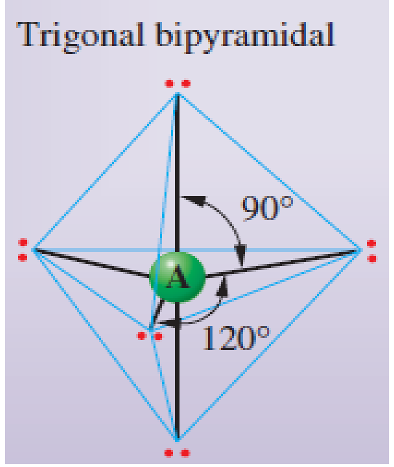

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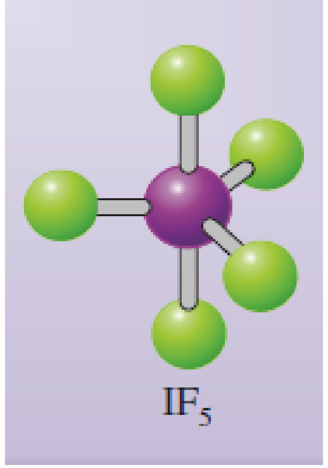

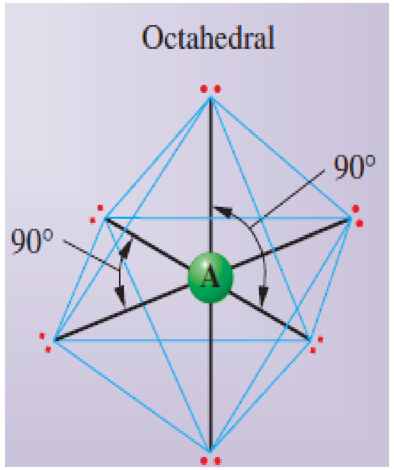

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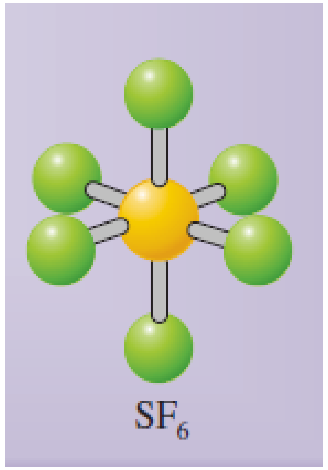
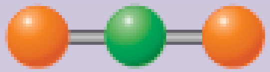
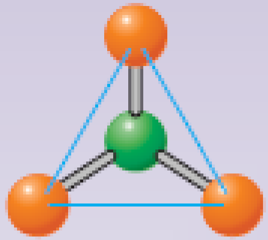
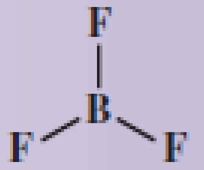
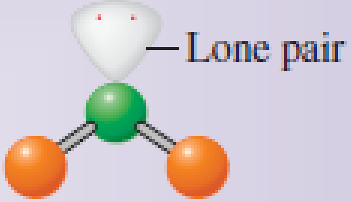
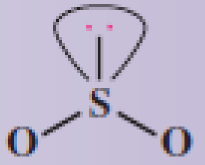
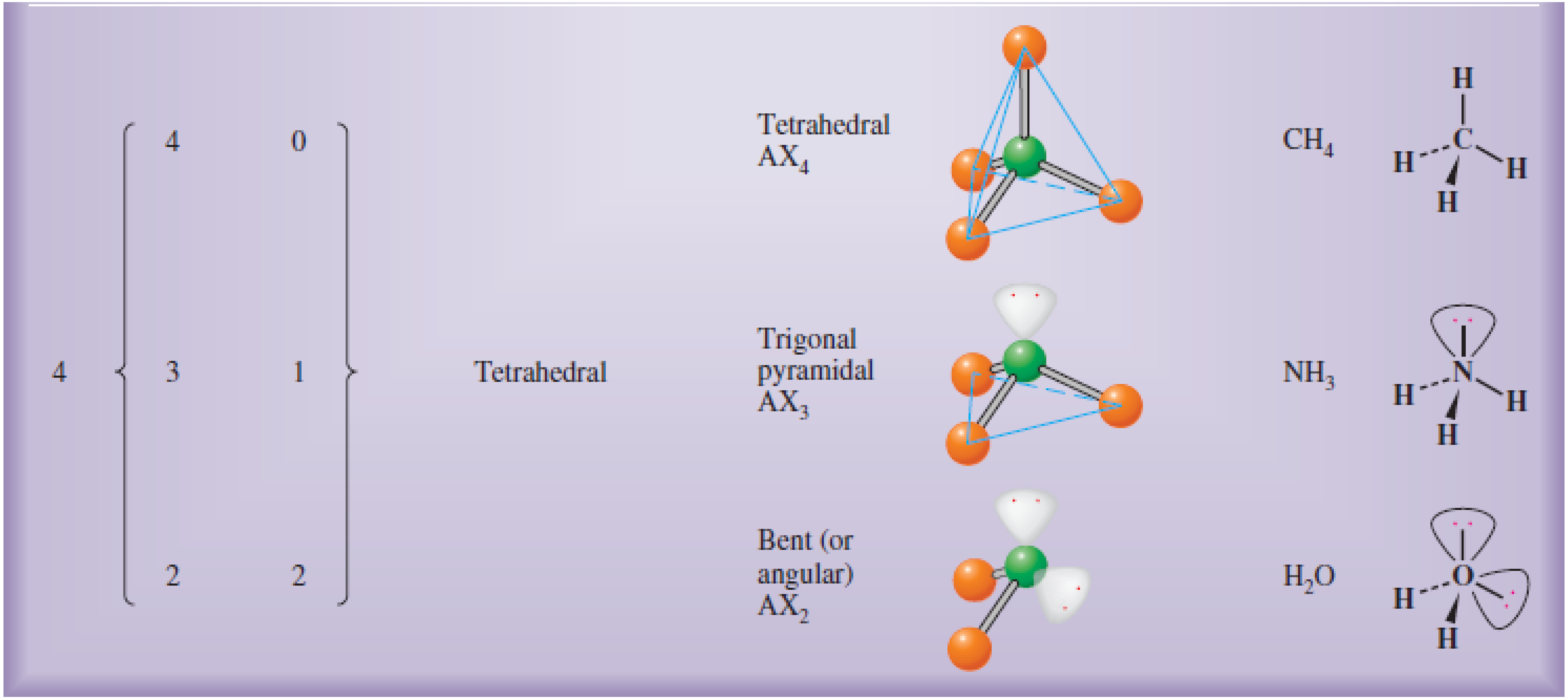


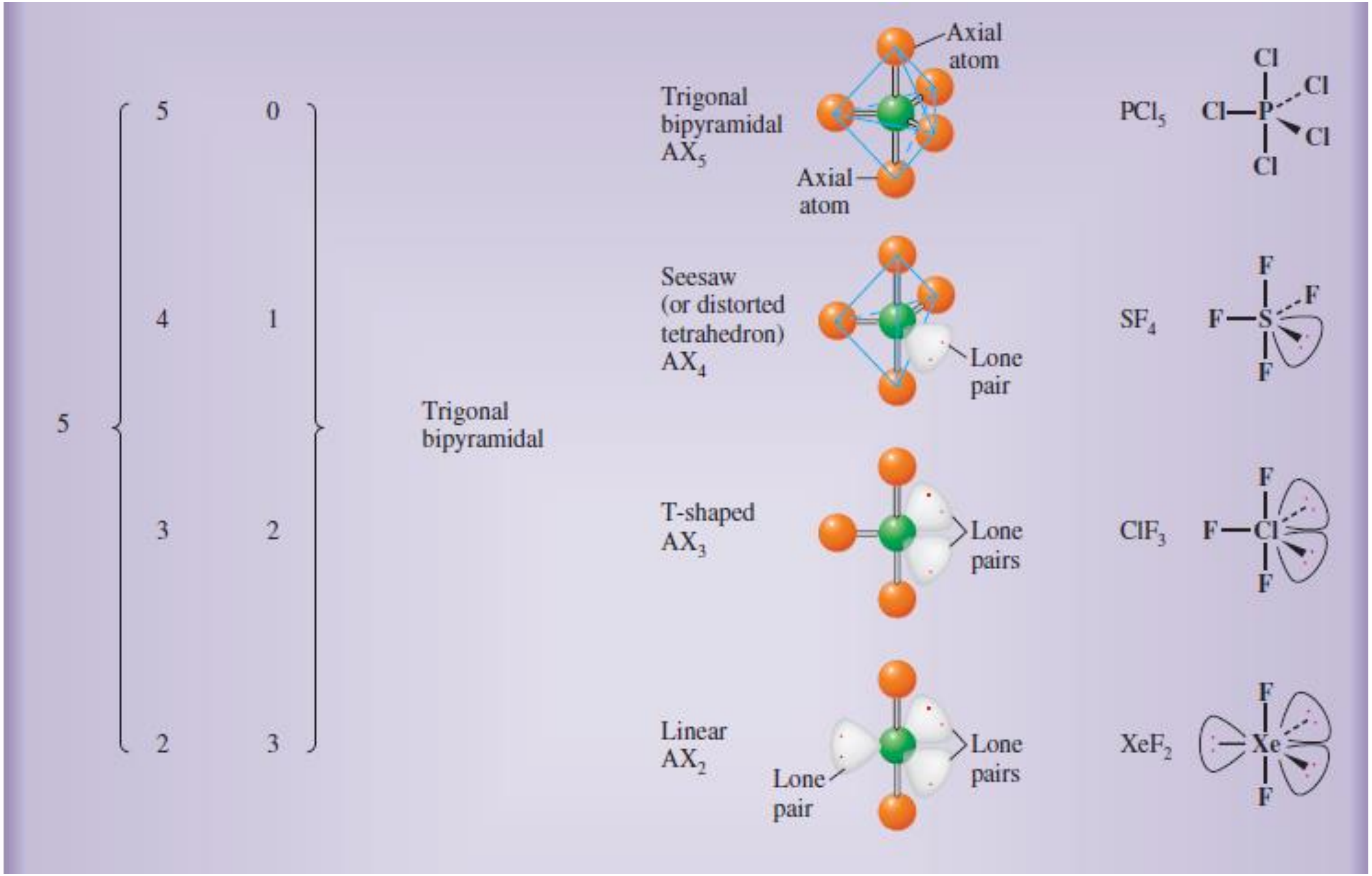
Valence Shell Electron Pair Repulsion Theory

Electron Pairs: Total	Electron Pairs: Bonding	Electron Pairs: Lone	Arrangement of Pairs	Molecular Geometry		Example	
2	2	0	Linear	Linear AX_2		BeF_2	F—Be—F
3	3	0	Trigonal planar	Trigonal planar AX_3		BF_3	
	2	1		Bent (or angular) AX_2		SO_2	

Valence Shell Electron Pair Repulsion Theory

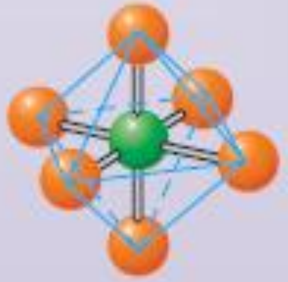
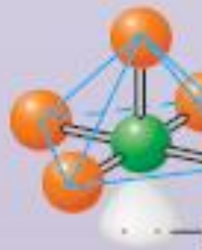
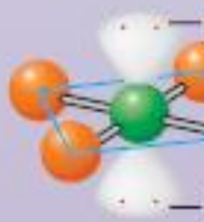


Valence Shell Electron Pair Repulsion Theory



Valence Shell Electron Pair Repulsion Theory

$H=1/2(V+X-C+A)$
$L=H-X-D$
Here, X= Monovalent atom number
D= Divalent atom number

6	6	0	Octahedral	Octahedral AX_6		SF_6	
	5	1		Square pyramidal AX_5	 Lone pair	IF_5	
	4	2		Square planar AX_4	 Lone pair, Lone pair	XeF_4	

Molecular geometry and polarity

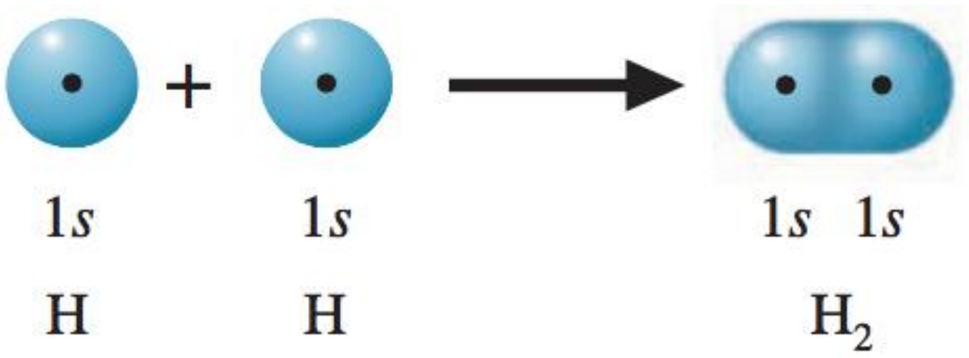
Valence Bond Theory

The main postulates of the valence bond theory are:

- A covalent bond is formed due to the overlap of the outermost half-filled orbitals of the combining atoms. The strength of the bond is determined by the extent of overlap.
- The two half-filled orbitals involved in the covalent bond formation should contain electrons with opposite spins. The two electrons then move under the influence of both the nuclei.
- The completely-filled orbitals (orbitals containing two paired electrons) do not take part in the bond formation.
- An *s*-orbital does not show any preference for direction. The non-spherical orbitals such as, *p*- and *d*-orbitals tend to form bonds in the direction of the maximum overlap, *i.e.*, along the orbital axis.
- Between the two orbitals of the same energy, the orbital which is non-spherical (*e.g.*, *p*- and *d*- orbitals) forms stronger bonds than the orbital which is spherically symmetrical, *e.g.*, *s*-orbital.

Valence Bond Theory

For example, consider the formation of the H_2 molecule from atoms. Each atom has the electron configuration $1s^1$. As the H atoms approach each other, their 1*s* orbitals begin to overlap and a covalent bond forms



Valence bond theory explains why two He atoms do not bond. Suppose two He atoms approach one another and their 1*s* orbitals begin to overlap. Each orbital is doubly occupied, so the sharing of electrons between atoms would place the four valence electrons from the two atoms in the same region. This, of course, could not happen. As the orbitals begin to overlap, each electron pair strongly repels the other. The atoms come together, then fly apart.

Valence Bond Theory : Orbital Overlap

Types of atomic orbital overlap leading to the formation of covalent bond:

1. s - s overlap

In this type of overlap, half-filled s -orbitals of the two combining atoms overlap each other.

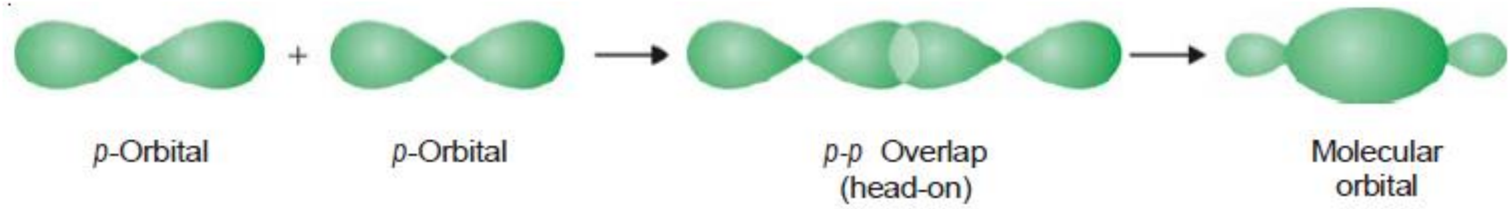
2. s - p overlap

Here a half-filled s -orbital of one atom overlaps with one of the p -orbitals having only one electron in it.

Valence Bond Theory : Orbital Overlap

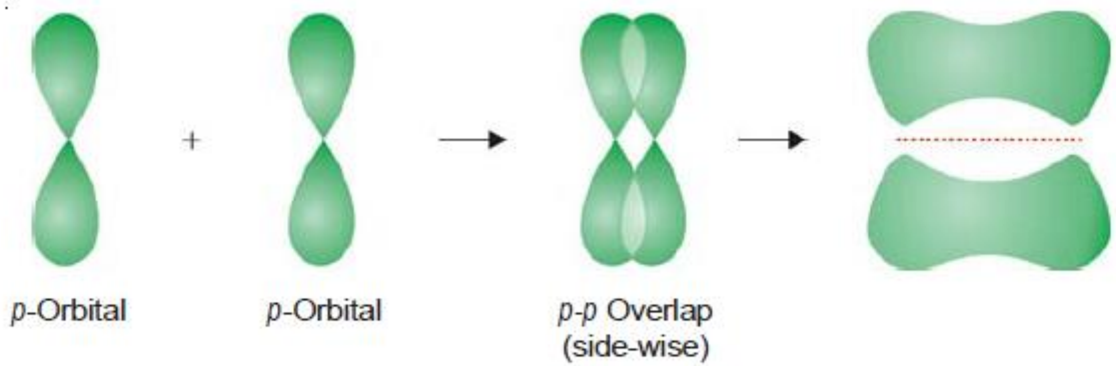
3. p-p head on overlap

This is called head on, end-on or end-to-end linear overlap. Here, the overlap of the two half-filled p-orbitals takes place along the line joining the two nuclei.



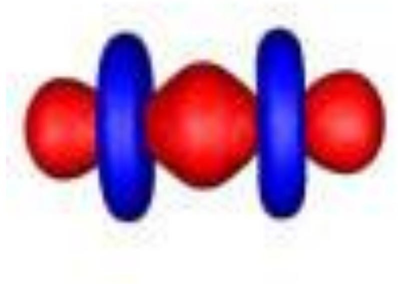
4. p-p sideways overlap

This is also called lateral overlap. In this types of overlap, two p-orbitals overlap each other along a line perpendicular to the internuclear axis, i.e., the two-overlapping p-orbitals are parallel to each other.

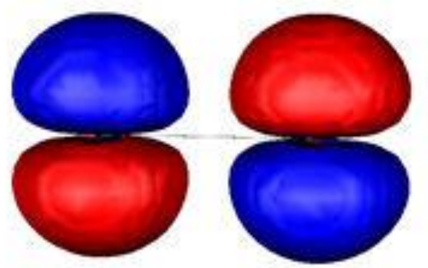


Sigma (σ) and Pi (π) Bond

A covalent bond formed due to the overlap of orbitals of the two atoms along the line joining the two nuclei (orbital axis) is called **sigma (σ) bond**. For example, the bond formed due to s-s and s-p, and p-p overlap along the orbital axis are sigma bonds.



A covalent bond formed between the two atoms due to the sideways overlap of their p -orbitals is called a **pi (π) bond.**



Sigma (σ) and Pi (π) Bond

Sigma (s) bond	Pi (π) bond
1. It is formed due to axial overlap of the two orbitals. The overlap may be of s-s, s-p, p-p orbitals.	**1.** This bond is formed by the lateral (sideways) overlap of two p-orbitals.
2. There can be only one **σ** bond between atoms.	**2.** There can be more than one π-bond between the two atoms.
3. The electron density is maximum and cylindrically symmetrical about the bond axis.	**3.** The electron density is high along a direction at right angle to the bond axis.
4. The bonding is relatively strong.	**4.** The bonding due to a π-bond is weak.
5. Free rotation of atoms about sigma (**σ**) bond is possible.	**5.** Free rotation about a π bond is not possible.
6. It can be formed independently, i.e., there can be a sigma (**σ**) bond without having a π bond in any molecule.	**6.** The π bond is formed only after **σ** bond has been formed.

Hybridization

The process of mixing of the atomic orbitals of nearly the same energy to produce equal number of new identical orbitals is called hybridization.

Atomic orbitals undergo hybridization only if the following conditions are satisfied:

- Atomic orbitals of the same atom participate in hybridization. Electrons present in these atomic orbitals do not participate in the hybridization and occupy the hybrid orbitals as usual.
- The atomic orbitals participating in hybridization should have nearly equal energy.

The characteristics of hybrid orbitals are:

- The number of hybrid orbitals formed is equal to the number of the atomic orbitals participating in hybridization.
- All hybrid orbitals are equivalent in shape, and energy, but different from the participating atomic orbitals.
- A hybrid orbital which takes part in the bond formation must contain only one electron in it.
- A hybrid orbital, like atomic orbitals, cannot have more than two electrons. The two electrons should have their spins paired.
- Due to the electronic repulsions between the hybrid orbitals, they tend to remain at the maximum distance from each other.

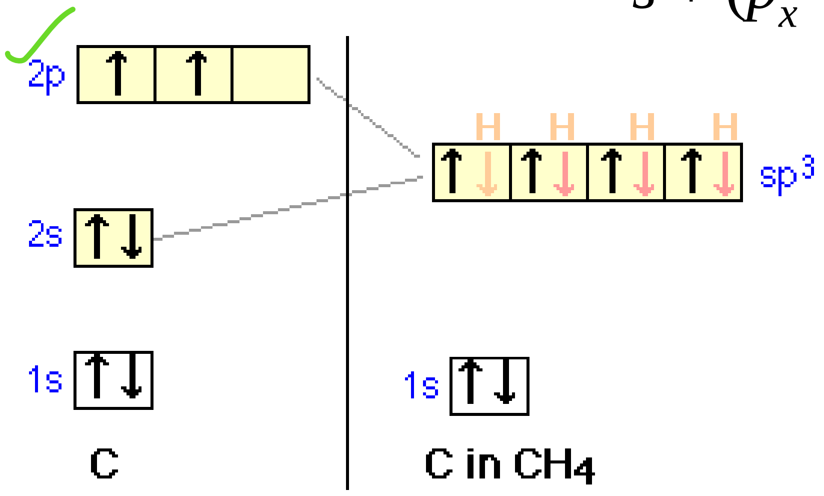
Types of Hybridization

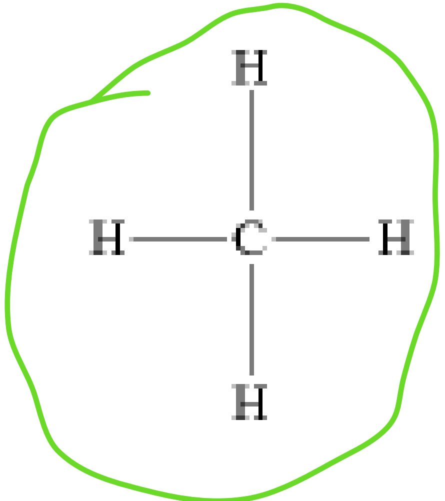
Depending upon the nature of the orbitals involved in hybridization, different types of hybridization are possible.

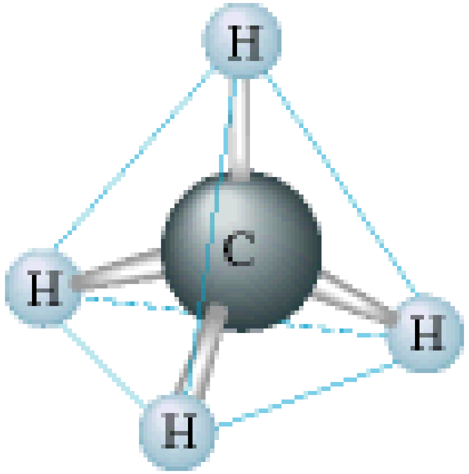
***sp*^{3} hybridization:**

In this type of hybridization mixing of one s-orbital and three p-orbitals of the valence shell to form four sp^3 hybrid orbital of equivalent energies and shape. There is 25% s-character and 75% p-character in each sp^3 hybrid orbital. The four sp^3 hybrid orbitals are directed towards the four corners of the tetrahedron. The angle between sp^3 hybrid orbital is 109.5°. These four orbitals undergo mixing to provide four new hybrid orbitals.

$$s + (p_x + p_y + p_z) \longrightarrow sp^3$$





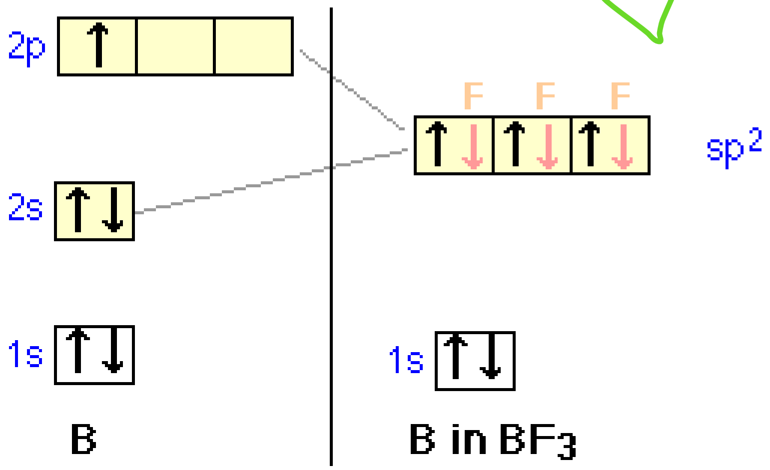


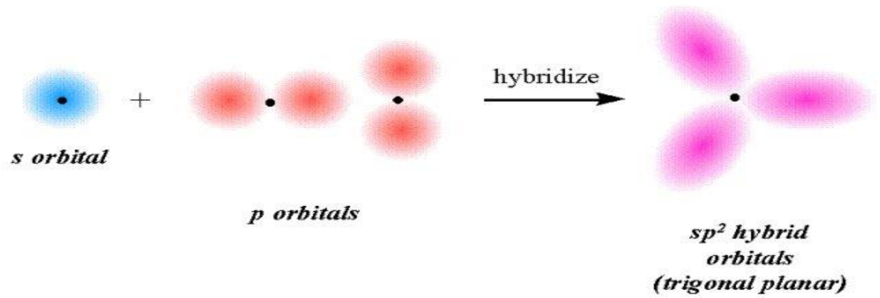
Types of Hybridization

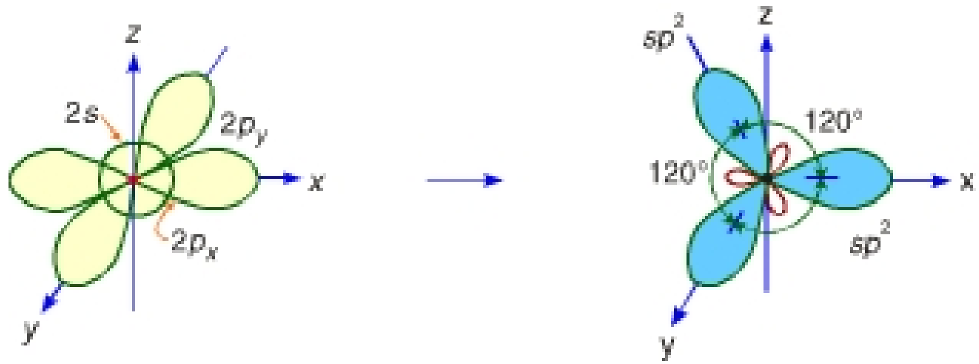
sp^2 hybridization:

One s and two p (say p_x and p_y) orbitals of an atom undergo mixing to produce three equivalent sp^2 hybridized orbitals. The three sp^2 hybrid orbitals are oriented in a plane along the three corners of an equilateral triangle, *i.e.*, they are inclined to each other at an angle of 120°. The third p-orbital (say p_z here) remains unchanged.

$s + (p_x + p_y) \rightarrow sp^2$





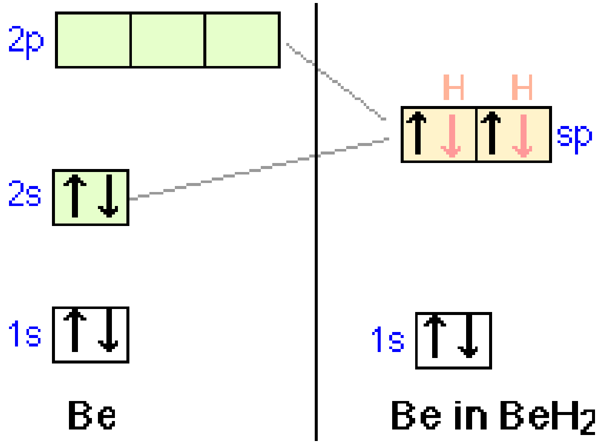


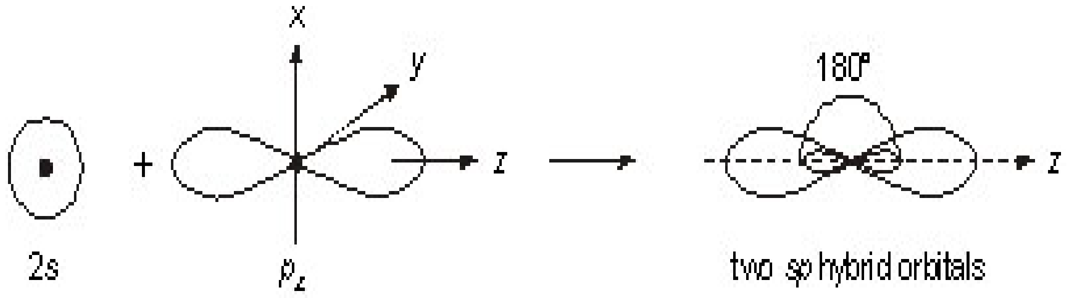
The pictorial representation of sp^2 hybridisation.

Types of Hybridization

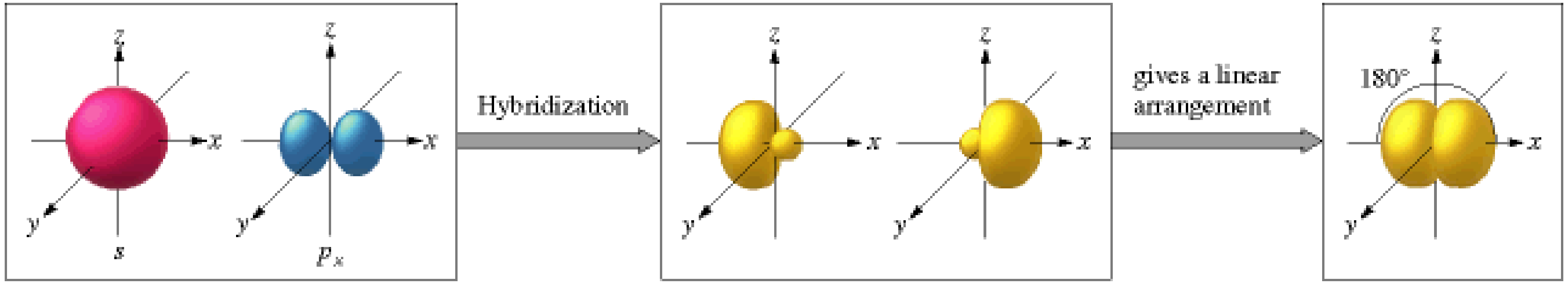
***sp* hybridization:**

In this type of hybridization, one *s* and one *p* (say p_z) orbitals belonging to the same main energy level hybridize to give two *sp* hybrid orbitals. These *sp* hybrid orbitals are oriented at an angle of 180° to each other. The other two *p*-orbitals (say $2p_x$ and $2p_y$) remain unhybridized and are oriented at right angles to each other and to the internuclear axis.



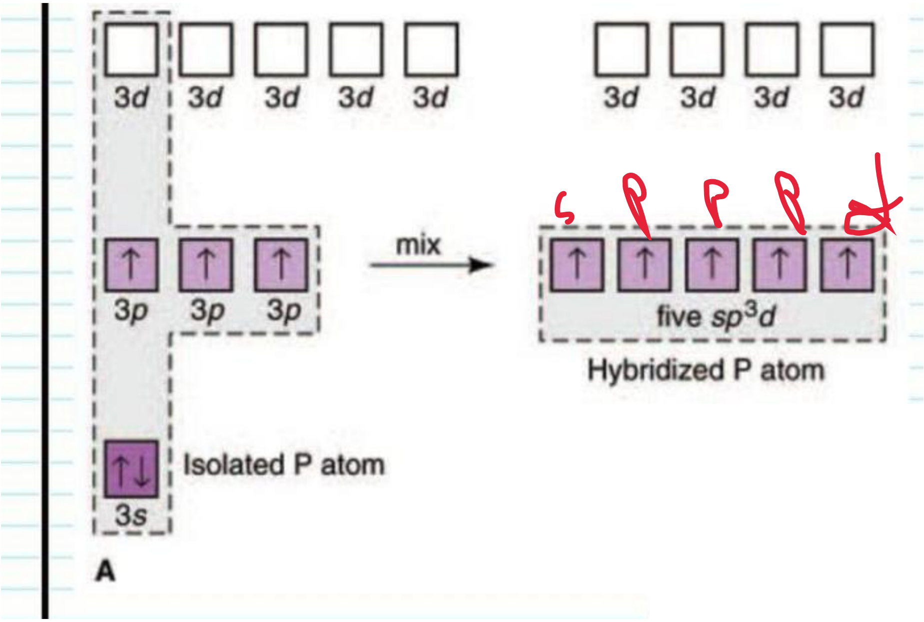


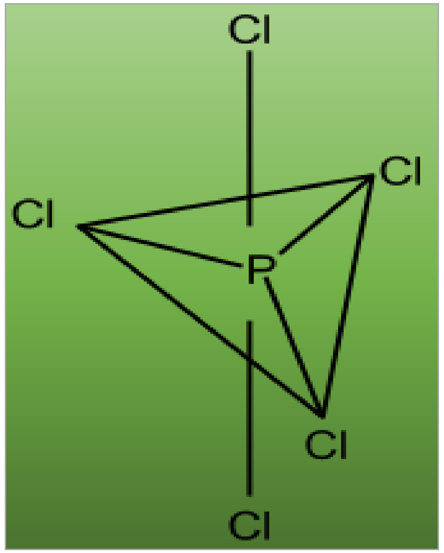
The *sp* hybridisation in carbon.

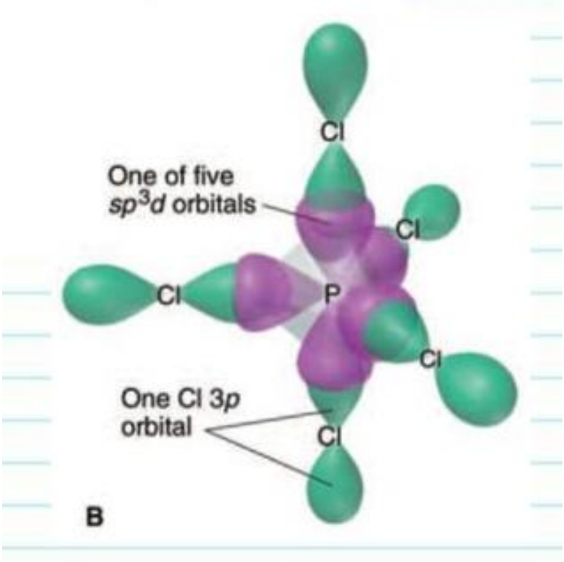


sp^3d hybridization:

One s, three p and one d orbitals hybridize to yield a set of five sp^3d hybrid orbitals which are directed towards the five corners of a trigonal bipyramidal. Three orbitals lie in one plane and make an angle of 120° with each other. The remaining two orbitals lying above and below the equatorial plane, make an angle of 90° with the plane.

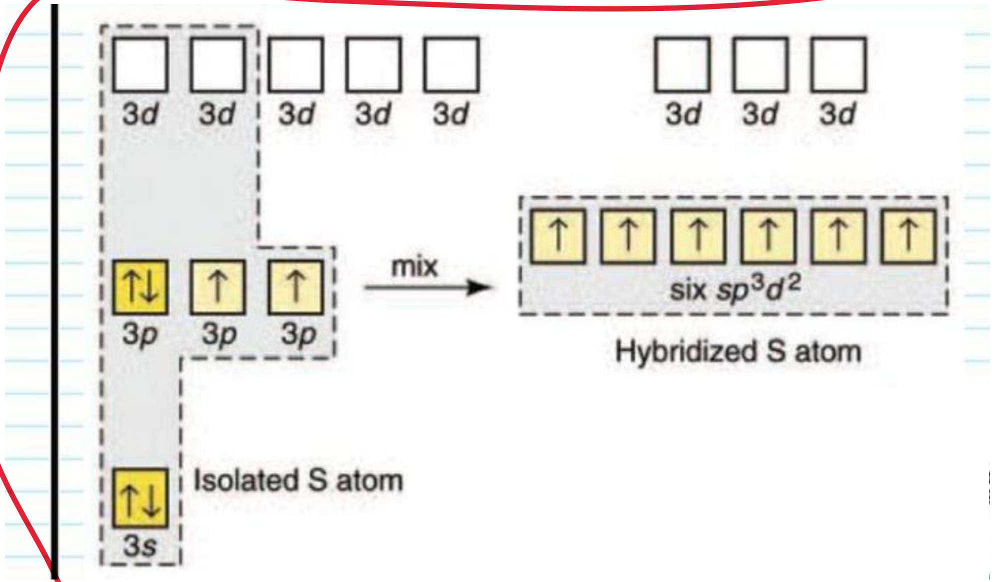


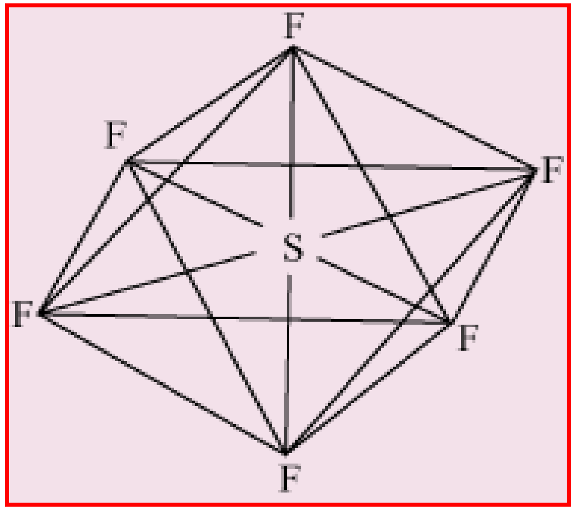


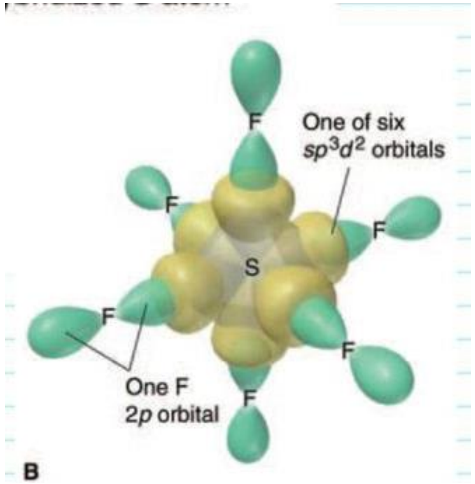


sp^3d^2 hybridization:

One s, three p and two d orbitals hybridize to form six new sp^3d^2 hybrid orbitals, which are projected towards the six corners of a regular octahedron.







Molecular Orbital Theory

Molecular orbital theory proposed by Hund and Mulliken in 1932 explains the formation of a covalent bond in a better way. According to molecular orbital theory all atomic orbitals of the atoms participating in molecule formation get disturbed when the concerned nuclei approach nearer. They all get mixed up to give rise to an equivalent number of new orbitals that belong to the molecule now. These are called molecular orbitals.

The main features of molecular orbital theory :

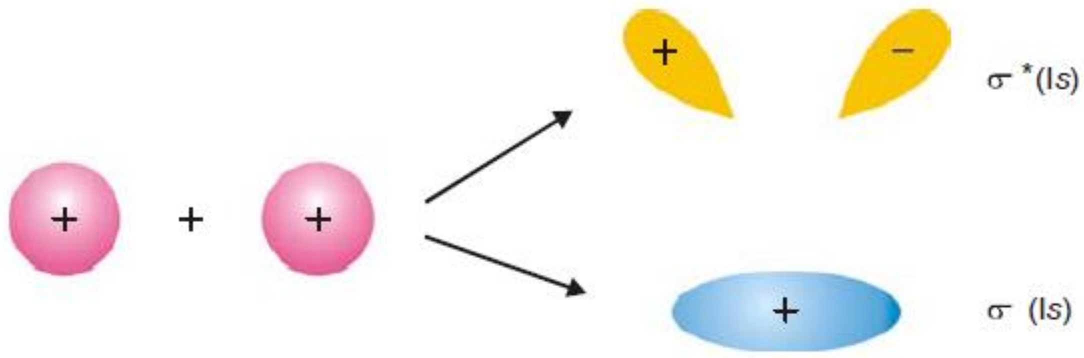
- A molecule is quite different from its constituent atoms. All the electrons belongs to the constituent atom are considered to be moving under the influence of all nuclei.
- Atomic orbitals of individual atoms combine to form molecular orbitals and these MOs are filled up in the same way as atomic orbitals are formed. In other words, Pauli’s exclusion principle, Aufbau principle and Hund’s rule of maximum multiplicity are followed.
- The molecular orbitals have definite energy levels.
- The shapes of MOs formed depend upon the shape of combining atomic orbitals.

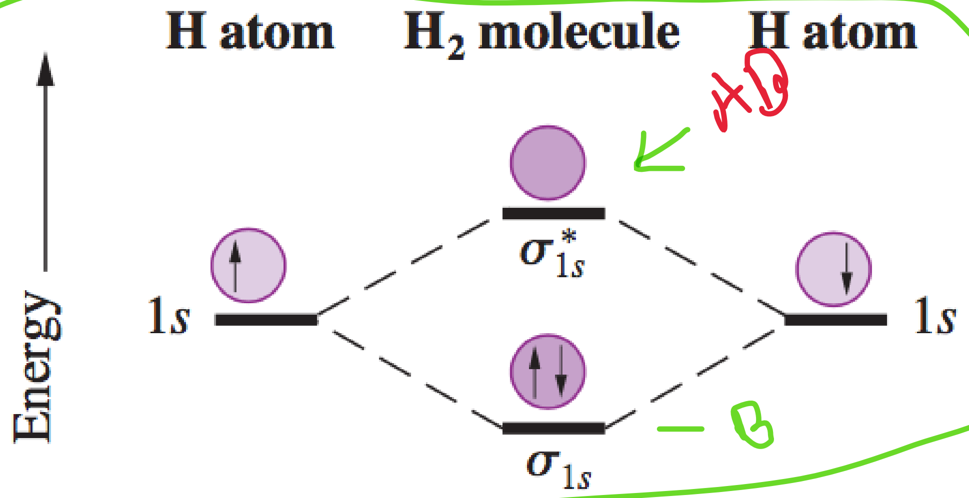
Molecular Orbital Theory

Each H atom has a 1s atomic orbital. When two H atoms come to a proper proximity, their 1s orbitals interact and produce two molecular orbitals: a bonding MO and an anti-bonding MO.

If the electrons are in phase, they have a constructive interference. This results in a bonding sigma MO (σ_{1s}). This MO has an increased probability of finding electrons in the bonding region.

If the electrons are out of phase, they have a destructive interference. This results in an anti-bonding sigma MO (σ^*_{1s}). This MO has a decreased probability of finding electrons in the bonding region.





Conditions for the Formation of MOs

Formation of MOs by the combination of atomic orbitals takes place only if the following conditions are satisfied:

(i) **The combining atomic orbitals should have nearly equal energies.** Only the atomic orbitals of nearly the same energy combine to form MOs. For example, *1s* atomic orbitals of two atoms can combine to form one bonding (σ *1s*) and one antibonding (σ^* *1s*) orbitals. The *1s* atomic orbital of one atom cannot combine with *2s* or *2p* atomic orbital of the other atom.

(*i*i) **The combining atomic orbitals should have the same symmetry.** The atomic orbitals are oriented in space. Only those atomic orbitals can combine to form molecular orbitals which have the same symmetry about the molecular axis. For example, a p_x orbital of an atom can combine with a p_x orbital of another atom. A p_x orbital cannot combine with a p_z orbital.

(*ii*i) **The combining atomic orbitals should overlap effectively.** MOs are formed only if the combining atomic orbitals overlap to a reasonable extent.

Conditions for the Formation of MOs

The presence of one or more unpaired electrons accounts for the **paramagnetic** nature of the molecule. The electronic configuration in which all the electrons are paired indicate the **diamagnetic** nature of the species.

The strength of a chemical bond is described in terms of a parameter called **bond order.**

Bond order $=(N_b-N_a)/2$

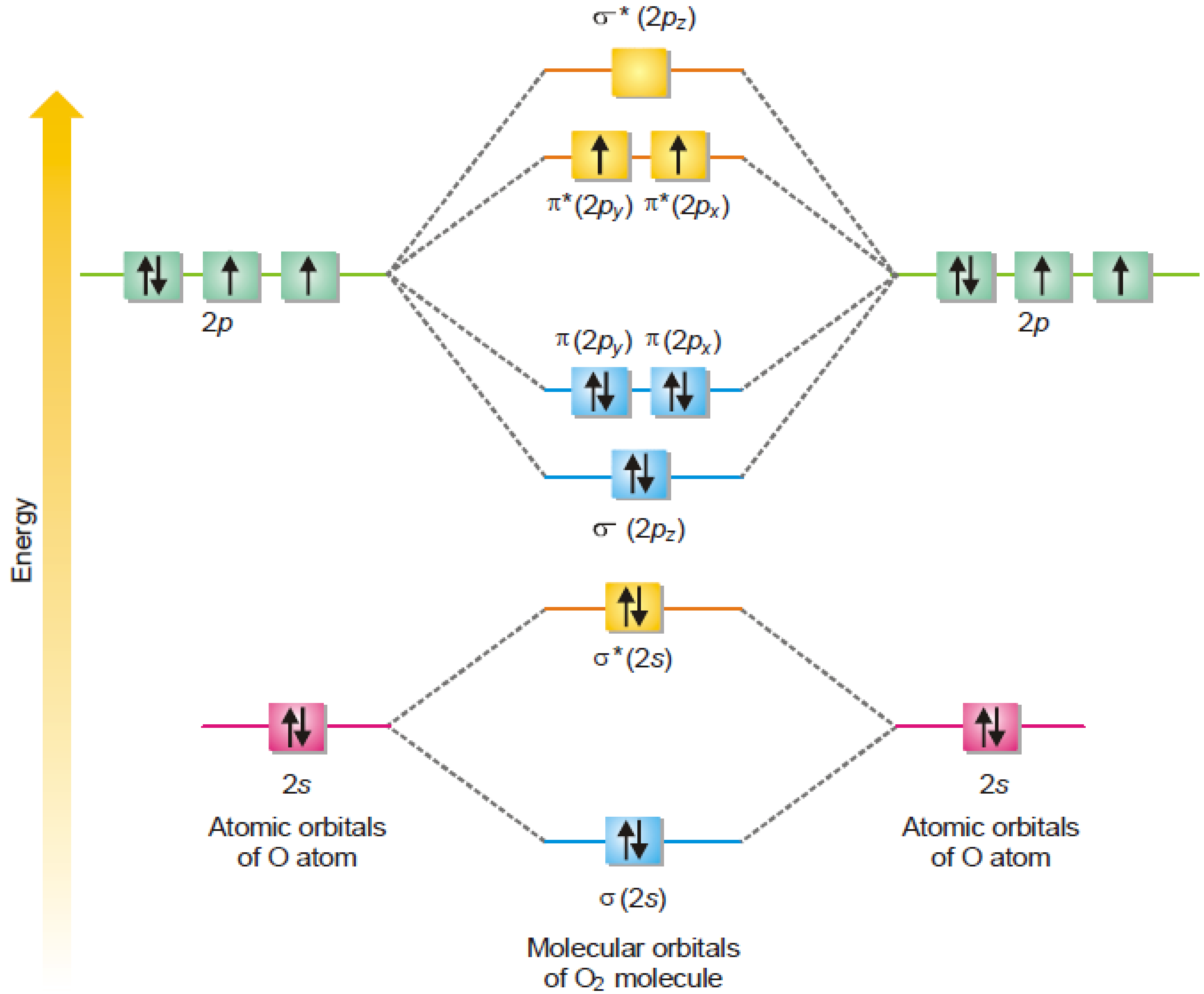
where, N_b is the total number of electrons in bonding MOs.

N_a is the total number of electrons in antibonding MOs.

(a) When, $N_b > N_a$: Bond order > 0 (+ve). →Stable bond.

(b) When, $N_b = N_a$: Bond order =0. → Unstable bond

Examples

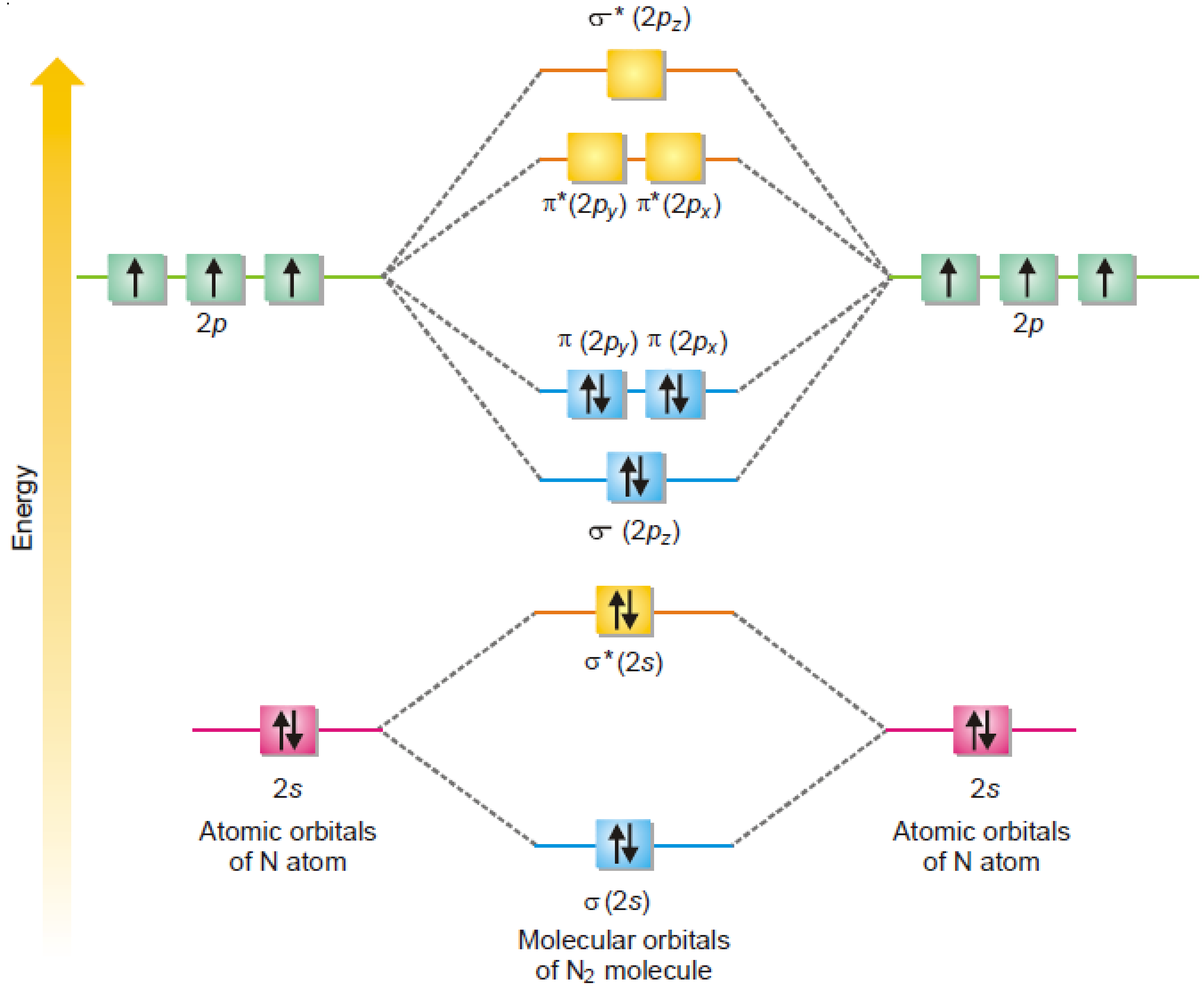


Electronic configuration:
$KK(\sigma_{2s})^2(\sigma^*_{2s})^2(\sigma_{2pz})^2(\boldsymbol{\pi}_{2px})^2(\boldsymbol{\pi}_{2py})^2\,(\boldsymbol{\pi}^*_{2px})^1\,(\boldsymbol{\pi}^*_{2py})^1(\sigma^*_{2pz})^0$

The ($1s$)^2($*1s$)^2 part of the configuration is often abbreviated KK

Bond order = (8-4)/2 = 2
There are two unpaired two electron on the antibonding orbital, therefore, O_2 is paramagnetic.

Examples

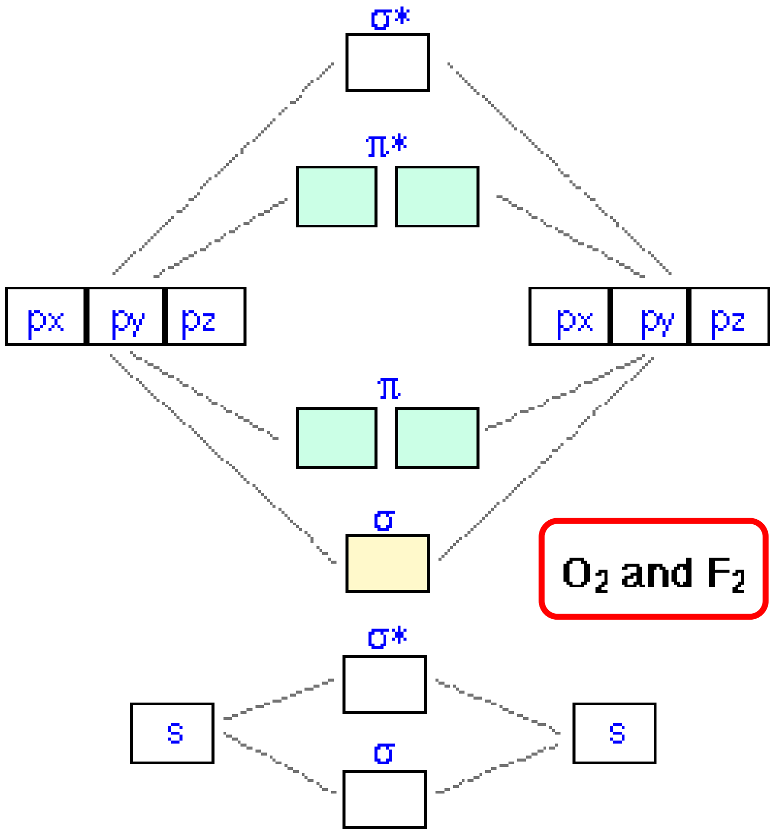


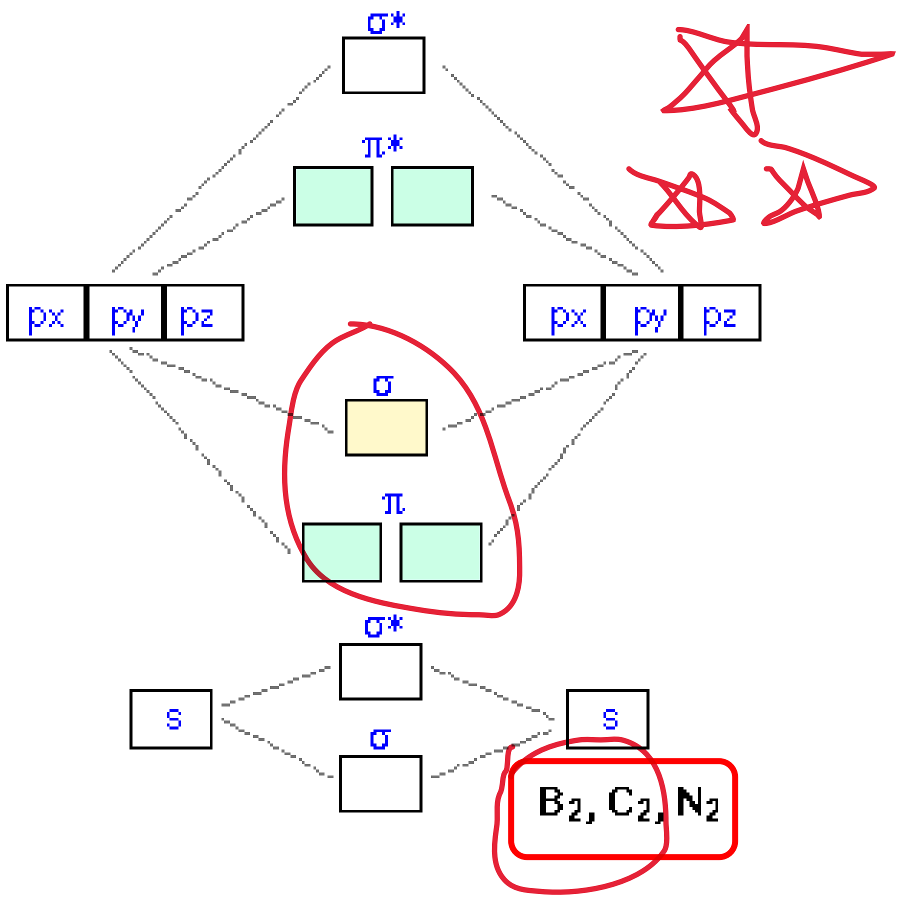
Electronic configuration:
KK$(\sigma_{2s})^2(\sigma^*_{2s})^2(\sigma_{2pz})^2(\boldsymbol{\pi}_{2px})^2(\boldsymbol{\pi}_{2py})^2\,(\boldsymbol{\pi}^*_{2px})^0\,(\boldsymbol{\pi}^*_{2py})^0(\sigma^*_{2pz})^0$

The (1*s*)^2(*1*s*)^2 part of the configuration is often abbreviated KK

Bond order = (8-2)/2 = 3
There are no unpaired electron on the antibonding orbital, therefore, N_2 is diamagnetic.

Examples





One minor complication that you should be aware of is that the relative energies of the s and p bonding molecular orbitals are reversed in some of the second-row diatomic.